
Optimising Vehicle Efficiency: *A Stochastic Approach to Minimising Energy Consumption on the Road*

Miguel Jeffrey Arias

School of Physics and Astronomy
University of Nottingham

May 15, 2024

Abstract

This project addresses the ongoing debate surrounding the most efficient and environmentally friendly driving practices in light of rising oil prices and the drive towards mitigating climate change effects around the globe. Employing an analytical approach, the study investigates optimal driving techniques for minimising energy consumption across varying terrain profiles and journey lengths. Through the use of a computational model and Monte Carlo optimisation method, the study simulates 300,000 journeys to determine the most effective strategies. Findings suggest that leveraging built up vehicle momentum and adapting driving strategies to certain terrain characteristics are key to fuel efficiency without compromising travel time. Moreover, anticipating and adjusting to upcoming terrain variations significantly contribute to energy-saving driving practices.

Contents

1	Introduction	4
2	Development & Performance of Physics Engine	4
2.1	Fundamental Assumptions & Model Requirements	4
2.2	Fiesta Performance Data	4
2.3	Physics Engine Block Model	7
2.4	Full Throttle Case Results	9
3	Monte Carlo Optimisation Method	13
3.1	A Theoretical Minimum	14
4	Results & Analysis	14
4.1	Validation Test	14
4.2	Number of Segments	15
4.3	Benchmark Test: Average Speeds	17
4.4	Road Length & Terrain Profiles	19
4.4.1	Flat	20
4.4.2	Incline	21
4.4.3	Decline	23
4.4.4	Flat Top	25
4.4.5	Flat Bottom	27
4.4.6	Bumpy	29
5	Conclusion	30
	References	32
	Appendix	33

1 Introduction

What is the most efficient way to drive a car? A question that many people, from urban commuters to lorry drivers, have been asking themselves as they travel to and from work or on a long-distance journeys ever since the invention and commercialisation of the internal combustion engine in 1860 by Étienne Lenoir[1].

Essentially, the objective of this project is split into two parts: first, defining a mathematical model that describes how a regular vehicle operates, which can be implemented in Python to simulate a real-world car journey; and secondly, using this model to figure out a general driving strategy for optimal fuel efficiency. It serves as a computational approach to finding either one optimal solution to our question, or more likely, a set of solutions with similar traits that are important for improving fuel economy on the road and how they vary with differing road profiles, time constraints, etc.

2 Development & Performance of Physics Engine

2.1 Fundamental Assumptions & Model Requirements

To solve this problem and ultimately determine the most efficient driving strategy, it is essential to make use of a model that realistically simulates a variety of vehicle journeys. Therefore, a computational model had to be developed, tailored specifically to the problem at hand. The program, written in Python due to it being the language that both participants knew the best, draws on the engineering principles behind internal combustion engines while maintaining clear and consistent general assumptions throughout. The analytical model simulates a car traveling along a two-dimensional road with a single forward direction, disregarding traffic, hazardous situations, and speed limits.

Inputs and outputs to the system are clearly defined, and the model answers important questions regarding the nature of the project, such as: What is to be varied, minimised and kept constant? What exactly constitutes a 'driving strategy' and what will the final solution look like? In this case, the solution will correspond to whichever driving strategy minimises the total energy used up by the modelled engine along a specific road profile.

For a manual car, there are three crucial actions that any driver can take to change how their car moves and how efficiently it drives: accelerate, shift gears and brake. Any form of braking would lead to a loss in generated power by the modelled engine and hence a pointless reduction in a journey's energy efficiency, meaning braking would never constitute as part of the most optimal driving strategy for such a simple model. Therefore, a braking element was omitted from the computer model. The RPM (revolutions per minute) range at which to upshift and downshift is kept constant throughout and set to upshift at 3500 and downshift at 1500 RPM, meaning the only aspect of manual driving that is studied in this project is the optimal throttle variation along a road would be and how it varies for a variety of factors such as average mean velocity, road distance and terrain profiles. This is, then, what the solution to the project is set to be: an optimal throttle variation along a certain road profile and keeping to a specified average speed.

In order to diagnose a journey's performance and test to see if the model works as expected, the Python model is able to return diagnostic information regarding velocity, position, gear changes, RPM, and torques as functions of time.

2.2 Fiesta Performance Data

A 2020 Ford Fiesta with a 1.0L EcoBoost engine (125hp and 6 speed manual) was modelled using data for its power and torque curves, along with data for the maximum velocities at redline RPM (redline

meaning the engine's rated maximum RPM, in this case 6500) for each gear individually. This data is showcased in Figures 1 and 2 where the values for max velocity at redline are given in Tables 1 and 2 displays the individual and overall gear ratios for each of the 6 gears.

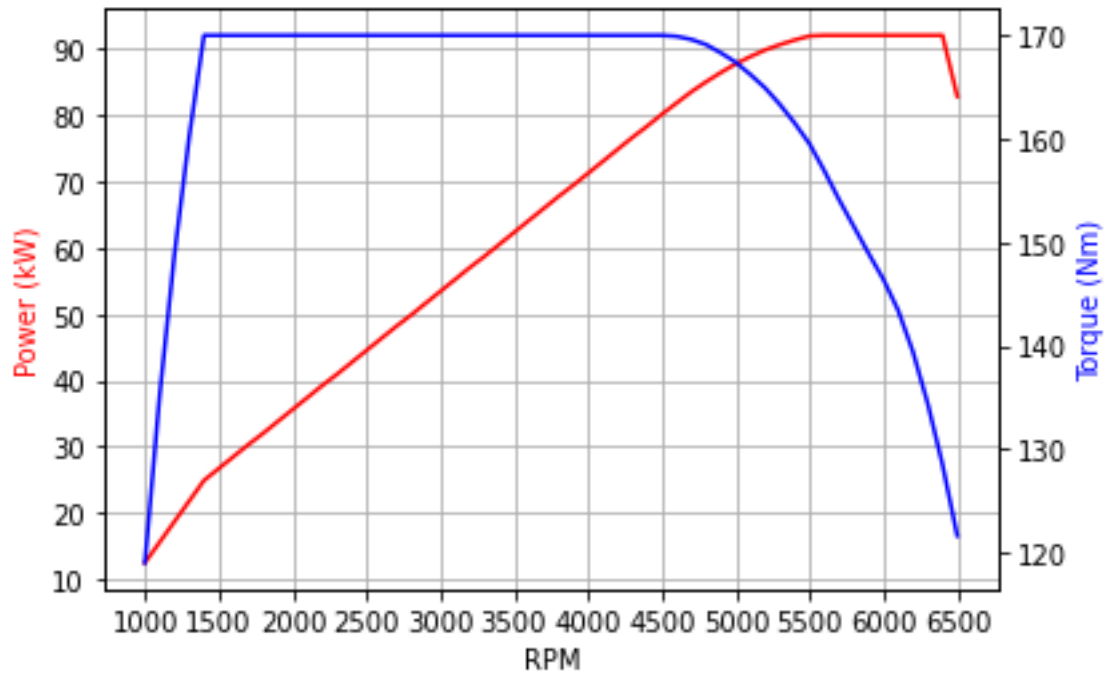


Figure 1: Power and torque curves for the 2020 Ford Fiesta used to model real-world values for a car's power output.

Table 1: Max speed on every gear.

Gear	Top Speed (mph)
1	35
2	62
3	94
4	127
5	158
6	189

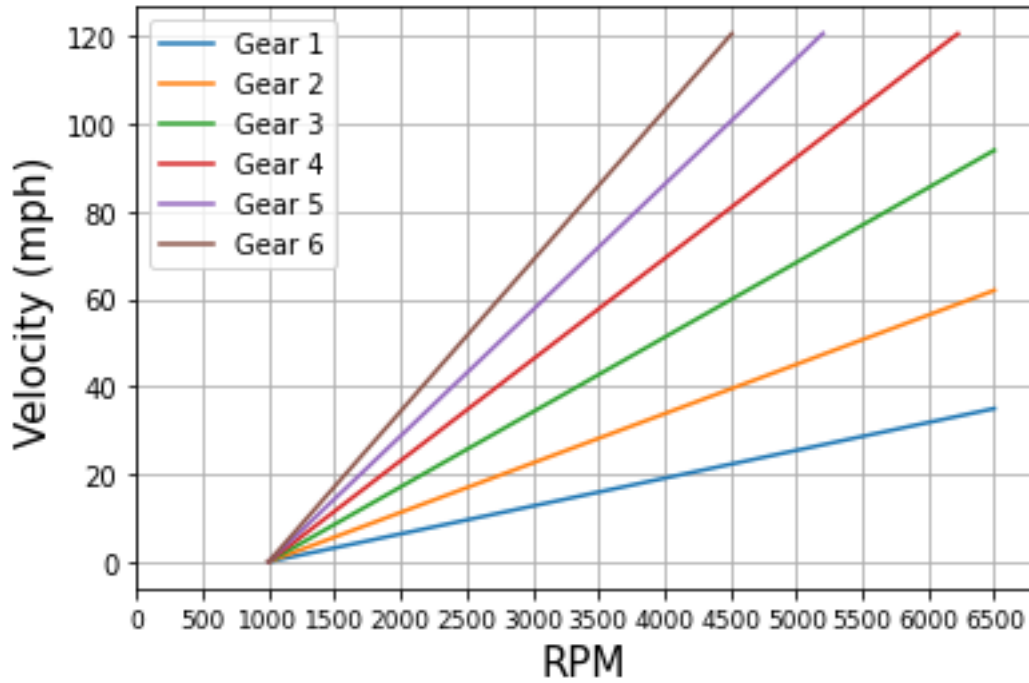


Figure 2: Relationship between velocity and RPM for each gear, assuming the vehicle begins to move from stationary with 1000 RPM.

Table 2: Gear ratios for every individual gear, overall ratio including the final drive gear ratio that is constant at 3.941.

Gear	Gear Ratio	Overall Ratio
1	3.417	13.47
2	1.958	7.72
3	1.276	5.03
4	0.943	3.72
5	0.757	2.98
6	0.634	2.5

All the data collected for this particular model of Ford Fiesta was obtained from the automobile catalog website [2], and can be seen in Table 3.

Table 3: List of values kept constant throughout the use of the physics engine in subsequent simulations.

Acceleration of Free Fall, g	$9.81ms^{-2}$
Vehicle Mass, M	$1279kg$
Drag Coefficient, C_d	0.321
Drag Area, A	$2.15m^2$
Tire Diameter, d_{tire}	$621mm$
Air Density, ρ_{air}	$1.2kgm^{-3}$
Rolling Friction Coefficient, μ_r (for 15-30 degrees)	0.03
Engine Thermal Efficiency, ν_{eng}	0.29
Drivetrain Transmission Efficiency, ν_{tr}	0.9
Radius of Crankshaft/Engine, R_{crank}	$5cm$

Figure 2 is not the same as the plot on the catalog website however, as in this case it is assumed that the modelled Fiesta starts moving with the engine at an RPM of 1000. The values for rolling friction coefficient, air density and typical values for vehicle efficiencies were found through brief investigation of papers and websites [3][4][5][6].

Please note that not all values used in the computer model are precisely accurate with the Ford Fiesta's measurements, such as the efficiencies or crank radius, however the selection of these values can be arbitrary to a certain degree, as long as the values match the order of magnitude of typical values found throughout engineering literature.

2.3 Physics Engine Block Model

The computational model consists of a set of functions and scripts written in Python that interact with each other to simulate car journeys where a distance, terrain profile and discrete number of segments are defined before each journey simulation. The discrete number of segments corresponds to how many sections the road is split into in order to define the number of throttle parameters along a certain road profile. In other words, this number defines the 'definition' of the resulting solution. Figure 3 displays a block diagram of how these functions work together to simulate the physical system in time.

Four functions are defined to compute how the vehicle accelerates in each time step within a 'while' loop set up to break once it has completed the designated journey, i.e. once the car's position along the road is equal to or just greater than the total length of the road.

The 'Engine' function computes the output engine torque as a function of throttle input, passes it through the 'Transmission' function to convert to wheel torque, and then subtracts the resistive forces (rolling friction, drag and slope weight).

From this net force, the acceleration is determined at every time step and so the new velocity and change in position along the road is recorded and used to plot any journey's performance with time. The driver function takes in the current RPM and decides when to upshift or downshift via an 'if' function, as any driver would do in real life by keeping an eye on their tachometer. Note that the "RPM Update" function uses the relation between velocity and RPM given in Figure 2 to update the engine's RPM as the velocity is updated by the "Wheels" function.

The road generator acts as a lookup for the slope of the road dependent on the car's position along the road. In this set of functions lies 6 options for different types of road profiles which have been defined for this project. These are the following:

- 'Flat': just a straight road with no variation in slope at any point
- 'Incline': slope of $+2^\circ$ for the entire length of the road

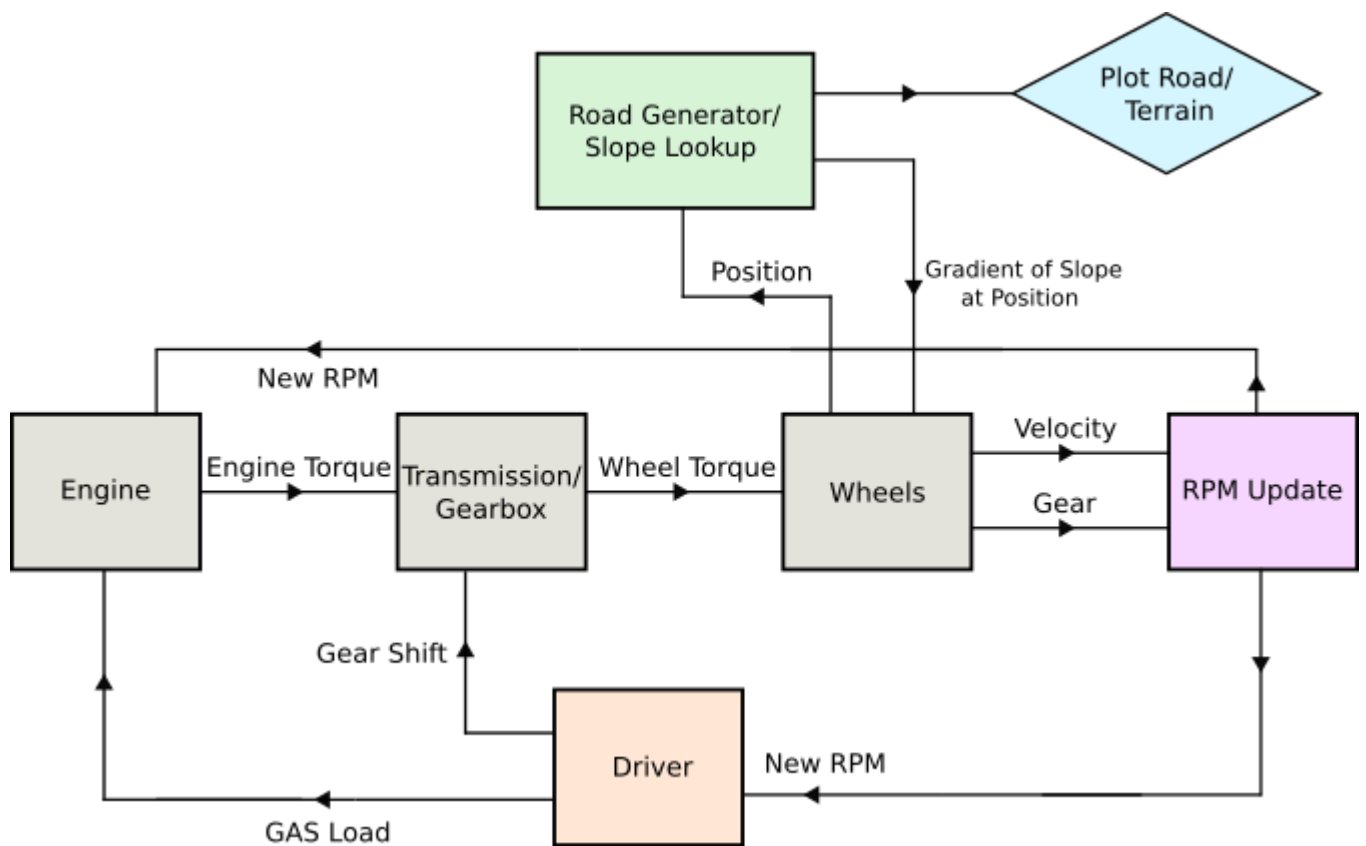


Figure 3: Block diagram showcasing the functionality of the Fiesta physics engine used throughout the entirety of this project.

- 'Decline': slope of -2° for the entire length of the road
- 'Flat Top': road split up into two equidistant sections, first one flat the second a slope of $+2^\circ$
- 'Flat Bottom': road split up into two equidistant sections, first one flat the second a slope of -2°
- 'Bumpy': an irregularly flat road with randomised slope angles between -3° and $+3^\circ$ randomly sampled from a distribution with a mean of 0 and following a seed for consistency in subsequent simulations.

2.4 Full Throttle Case Results

In this section, the Fiesta model is tested for the full throttle case along a 'Flat', half mile road.

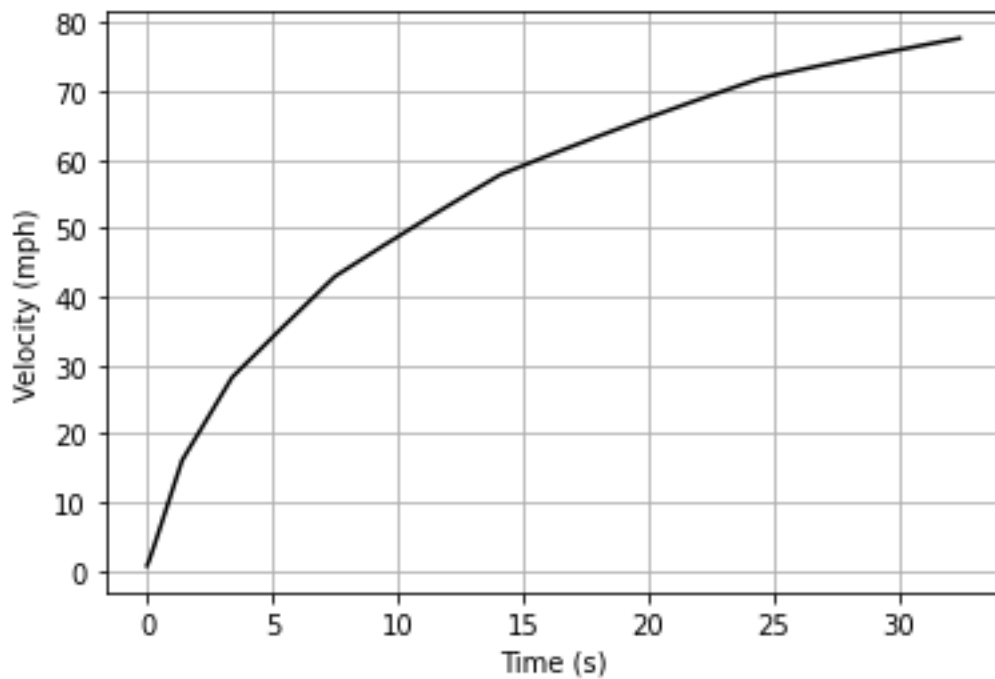


Figure 4: The velocity of the vehicle and how it evolves with time at full throttle along a $\frac{1}{2}$ -mile journey on a flat road.

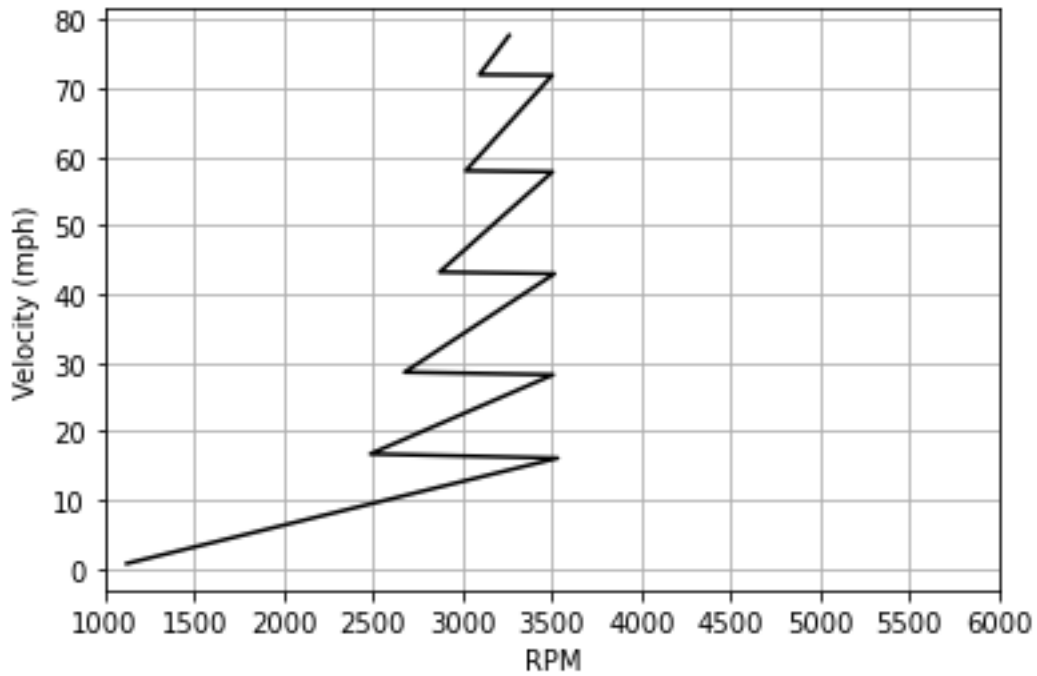


Figure 5: Graph plot displaying the gear changes made in terms of velocity against RPM.

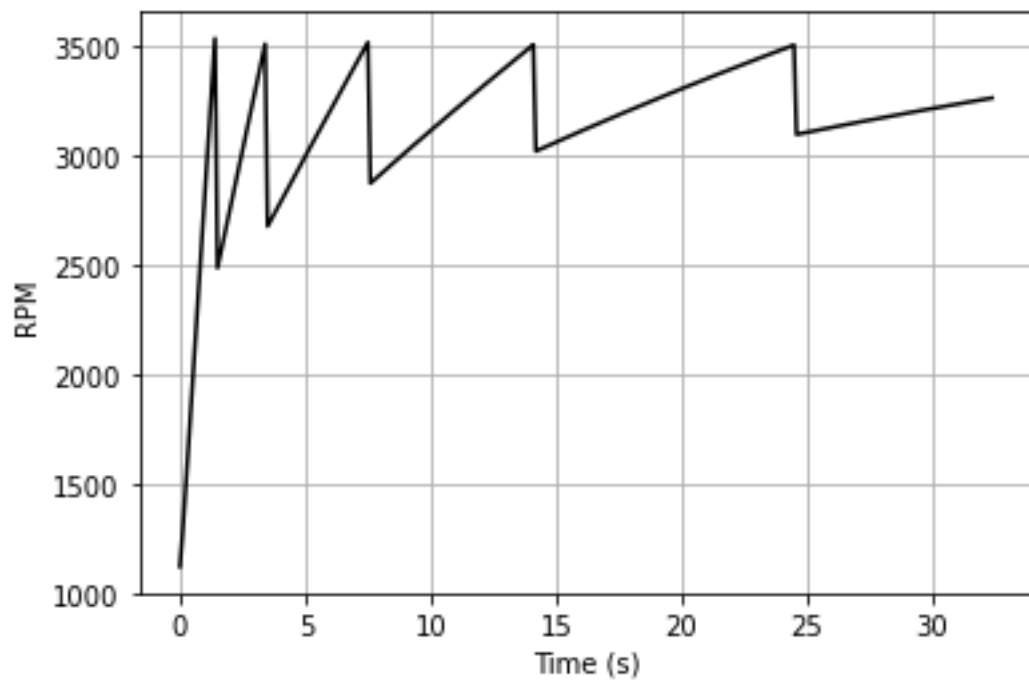


Figure 6: RPM evolution with time for full throttle along a flat, $\frac{1}{2}$ -mile road, indicating the different gear changes.

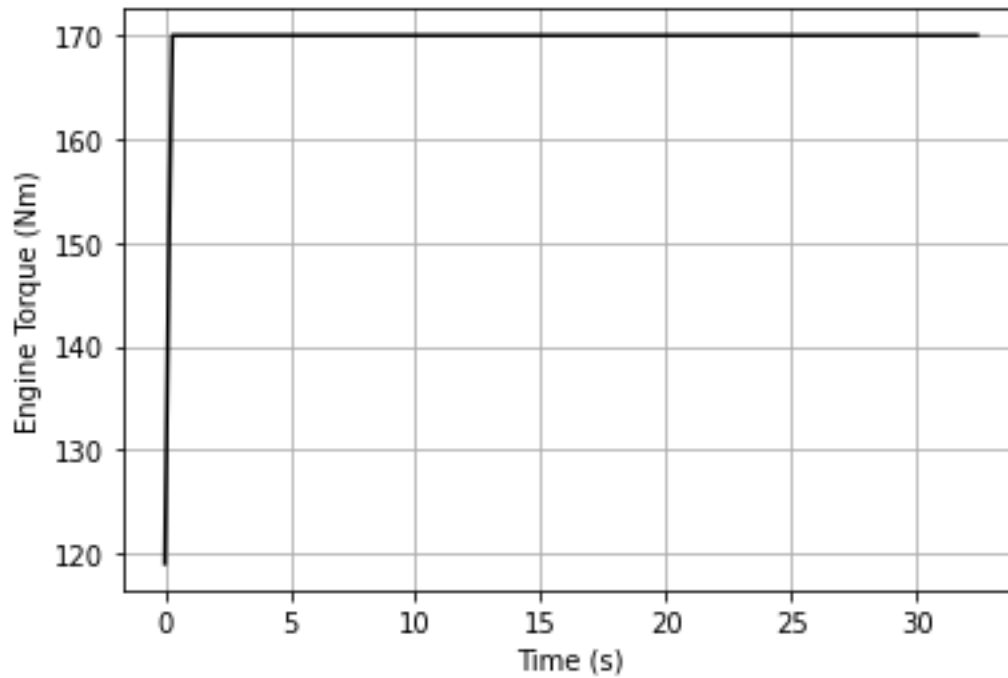


Figure 7: Output engine torque is at a constant 170Nm output for the entire journey due to the plateau nature of the EcoBoost's torque output, as well as the RPM shifting range being set to 1500 to 3500.

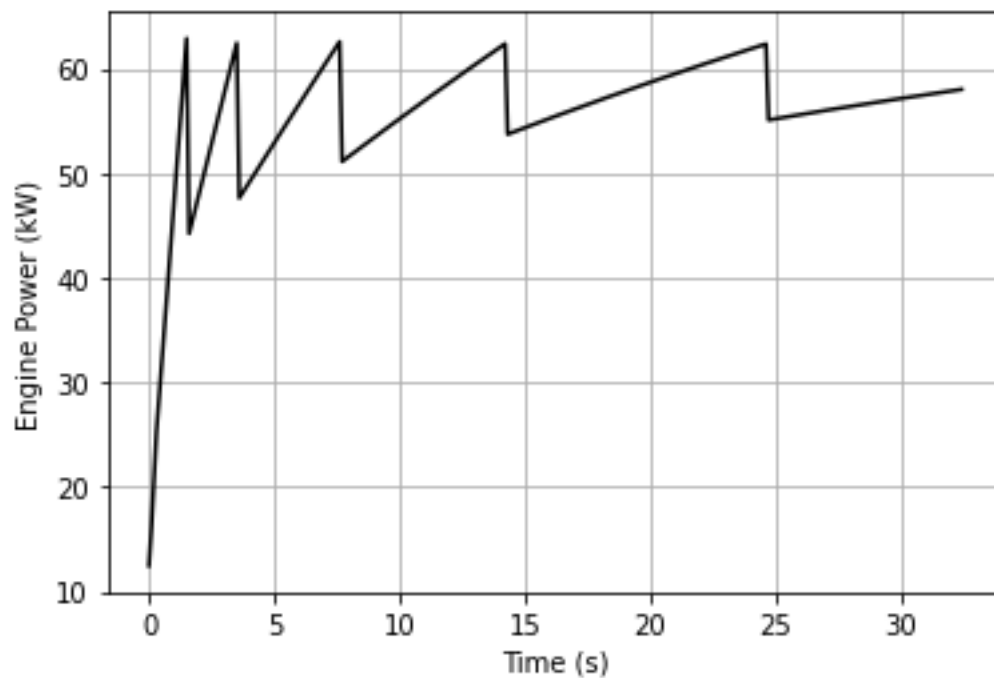


Figure 8: Engine power output over time for the flat, $\frac{1}{2}$ -mile road at full throttle.

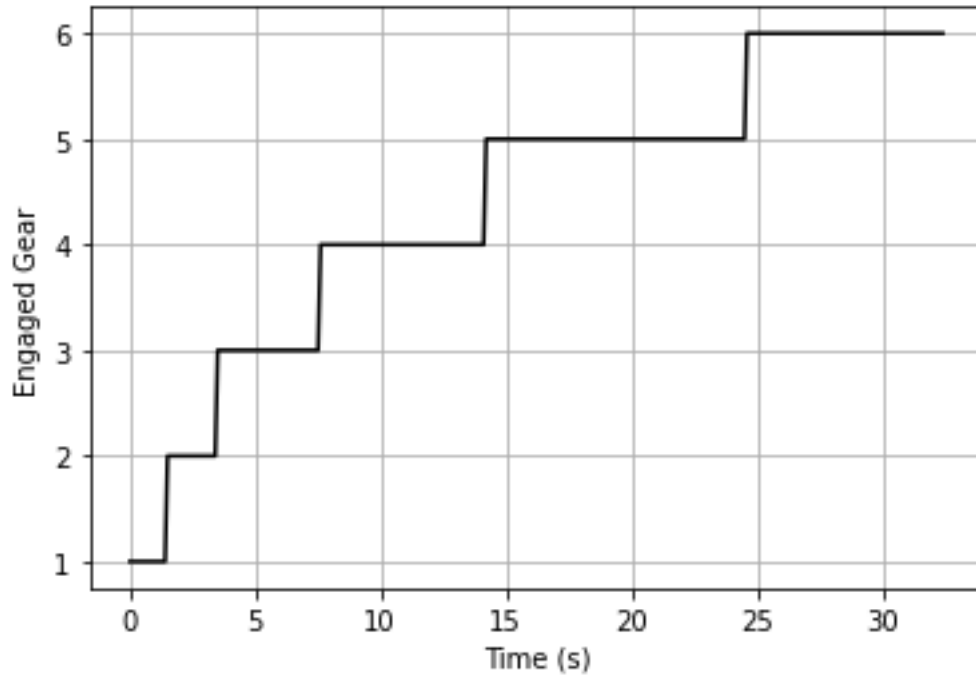


Figure 9: Plot of engaged gear with time for the full throttle journey along the flat, $\frac{1}{2}$ -mile road.

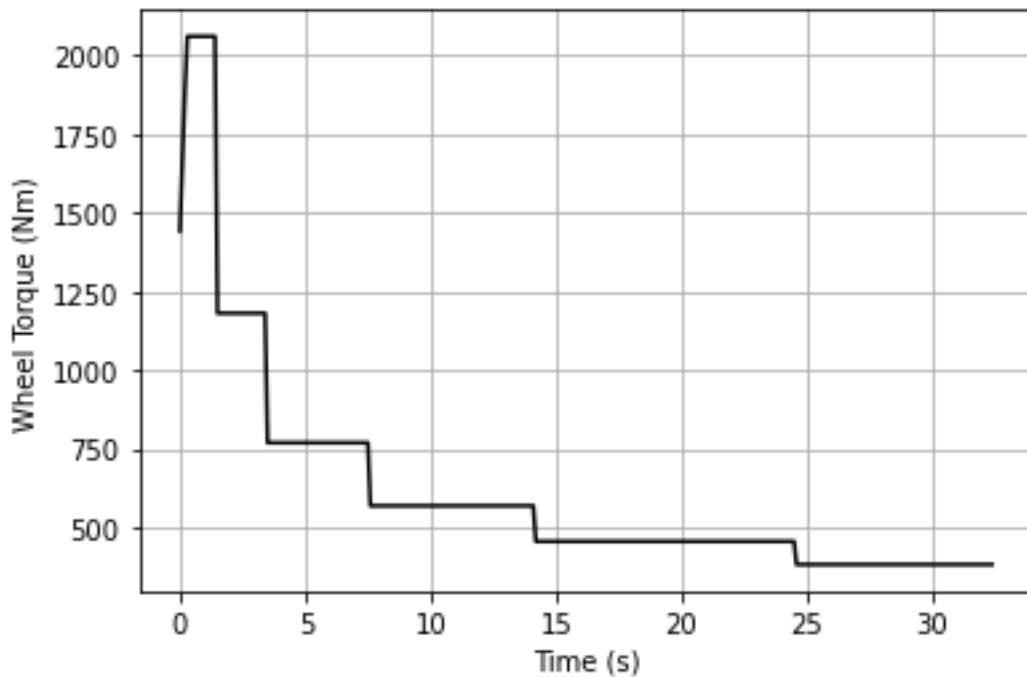


Figure 10: Showcases the torque output at the wheels over time for the full throttle journey along the flat, $\frac{1}{2}$ -mile road.

The Figures 4 to 10 showcase the performance of the physics engine for this full throttle case sce-

nario(max power & torque output at any given engine RPM). A half mile journey was simulated in order to display the performance reasonably clearly, as after a long distance the Fiesta travels at a somewhat constant velocity in 6th gear for the majority of the journey, making key features indistinguishable.

Each 'zig-zag' in Figure 5 depicts an upshift to a higher gear. Note that for different throttle variations across the length of the road, downshifting can also take place when the driver is not driving at full throttle and is reducing their speed, however in this case the car model never slows down and so never downshifts. This journey took the model Fiesta 33 seconds to complete and used a total of 0.50kWh of energy at the engine.

3 Monte Carlo Optimisation Method

Once a reliable and robust Python model for the car had been developed, all that needed to be done was to optimise it to find the most energy efficient way to drive it.

Deciding on an approach proved tricky, however with each attempt, from discretised throttle parameters within a nested 'for' loops to heuristic and machine learning algorithms such as the Nelder Mead and Differential Evolution algorithm, the project was understood better and better.

The main issue with the algorithm attempts of Nelder Mead and Differential Evolution is that they only minimise energy and output one result, and do not take into account the time taken to complete the journeys. After this realisation was made, it was clear to see that there needed to be a balance between energy-efficiency and time-efficiency, as the lowest possible energy is intuitively the journey that takes the longest and uses the absolute minimum energy to overcome resistive forces (not the most practical solution, as nobody is expected to take 3 hours to complete a 10 mile journey for example).

In order set up a suitable approach, it is necessary to first study and understand the relation between energy and time for many different journeys. Using a stochastic, random sampling technique whereby a random number between 0 and 1 is generated for each segment corresponding to the throttle parameter value at each corresponding segment for a particular journey, a set number of simulations were run to yield energy and time values along with the corresponding throttle parameters. This in essence is a 'Monte Carlo' optimisation technique as a solution will be determined after repeatedly sampling the throttle parameters from a probability distribution in order to plot the energy used by the engine against the time taken to complete any particular journey.

It is crucial to set a plan moving forward to test how our final solution will vary with certain factors. The factors which have been varied when collecting results comes down to the following list:

- i. Average velocity (target time)
- ii. Number of discrete segments along the road
- iii. Road length
- iv. Terrain profile

To investigate how each of these factors affect the optimal throttle variation, a benchmark road profile is defined as the 'Flat', 10 mile road, i.e. a long distance journey. Data for this benchmark profile as well as various other profiles was collected for 300,000 journey simulations, individually. A total of 36 road profiles were tested and therefore a total of 10.8 million simulations were run in order to obtain results for this project.

Two separate programs were written for the collection and analysis of this data: One to acquire the data and store it as a CSV (comma separated value) file with a unique name depending on the road profile. For each file, energy, time for completion and individual throttle parameters were recorded for all the successful journeys out of the 300,000. The other program to analyse the individual CSV databases

by averaging the 100 throttle variations with the lowest energy outputs for a given window around any specified average velocity.

3.1 A Theoretical Minimum

For any road profile, there must be a minimum energy, E_0 , required to complete a journey along such road. Consequentially, it can be predicted that plots for energy against time in the following results section should approach this value of minimum energy as the journey approaches an infinite amount of time, $\lim_{t \rightarrow \infty}$. Following this line of reasoning, energy should follow a relationship like the one given in equation (1),

$$E(t) = E_0 + \frac{k}{(t)^2} \quad (1)$$

where t is the time taken to complete a journey, and k is some constant of proportionality dependent on the road profile.

To set up a validation test, the minimum energy required to complete the benchmark profile journey can be calculated mathematically first, then determined numerically from a curve fit in the results & analysis section. This minimum energy value, E_0 , corresponds to the energy required to overcome resistive forces, and therefore can be calculated quite simply from the equations used in the Fiesta model.

Calculating the minimum resistive rolling friction force, assuming no drag as $\lim_{v \rightarrow 0}$ (no velocity when journey takes infinite amount of time) and no slope for flat benchmark, a minimum energy at the wheels for 10 mile journey is calculated as $E = 1.68$ kWh. Now, assuming that this car stays in first gear, a conversion factor can be calculated to find the corresponding energy at the engine from energy used up at the wheels. Using the efficiency values, crank and tire radii as well as overall gear ratio in first found in Table 3, the minimum energy to be used up by the engine for the benchmark profile is calculated to be $E_0 = 2.97$ kWh.

4 Results & Analysis

This section will cover the results of the benchmark test for different average velocities as well as how the optimal driving strategy changes for the number of segments for which the road is divided into, an increase in the length of the road and a variation in different terrain/road profiles. It will also serve as a discussion & analysis as a way to interpret the trends observed in each plot as they are presented and what they may indicate as an overall energy efficient driving strategy.

4.1 Validation Test

First off, let's see how well the mathematical prediction went for the minimum required energy, E_0 .

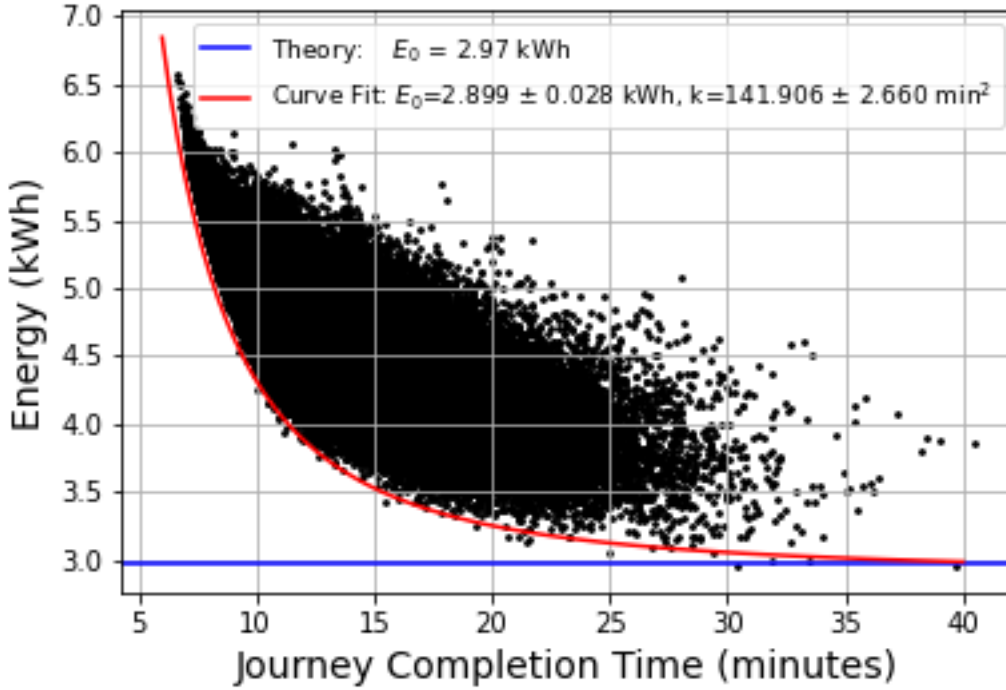


Figure 11: Scatter plot of successful individual journeys for benchmark profile, with a curve fitted using equation (1) to outline the boundary of successful driving strategies.

A general conclusion can be made by looking at the results of Figure 11, that there seems to be a clear boundary in the scatter plot separating successful and unsuccessful journeys, where the most energy efficient journeys lie on the boundary. This boundary is highlighted through the use of a curve fit approximated using the SciPy library function and equation (3). It is a rough fit, only used to visualise the plot in a cleaner way and in order to determine a rough value for the minimum energy E_0 . The fit calculates a value of $E_0 = 2.899 \pm 0.028 \text{ kWh}$ and $k = 141.906 \pm 2.660 \text{ min}^2$ for the constants in equation (1). Given that the value calculated based off of the mathematical model was determined to be $E_0 = 2.97 \text{ kWh}$, the model and the Monte Carlo simulations can be said to agree with each other, and so the optimisation method validated.

4.2 Number of Segments

In the following sections including this one, plots for the optimal throttle variations were obtained by finding the average of 100 throttle combinations with the lowest energy, at a specified average velocity or target time (these two terms can be used interchangeably), for the given road profile. The raw scatter plots (no curve fits) for each road profile used in the same order as used in the following sections are showcased in the appendix.

For this section, the number of segments are varied for a 'Flat', 10 mile road, so that the definition of the solution can be contrasted.

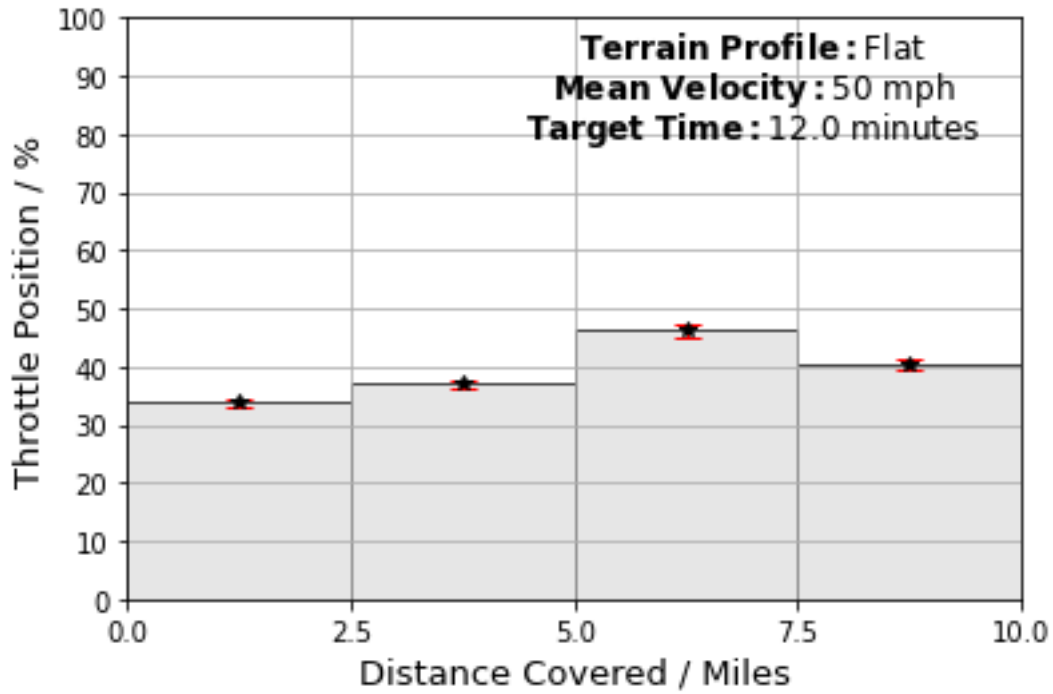


Figure 12: Plot of optimal throttle variation for a car travelling along the 'Flat', 10 mile road, split into 4 segments, at an average speed of 50 mph.

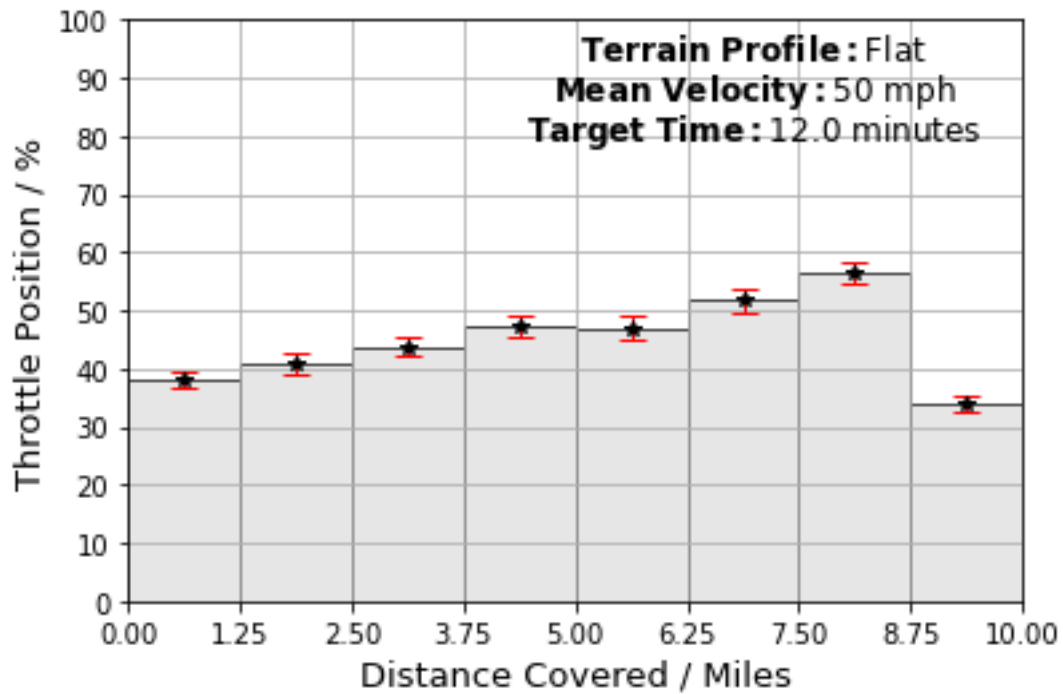


Figure 13: Plot of optimal throttle variation for a car travelling along the 'Flat', 10 mile road, split into 8 segments, at an average speed of 50 mph.

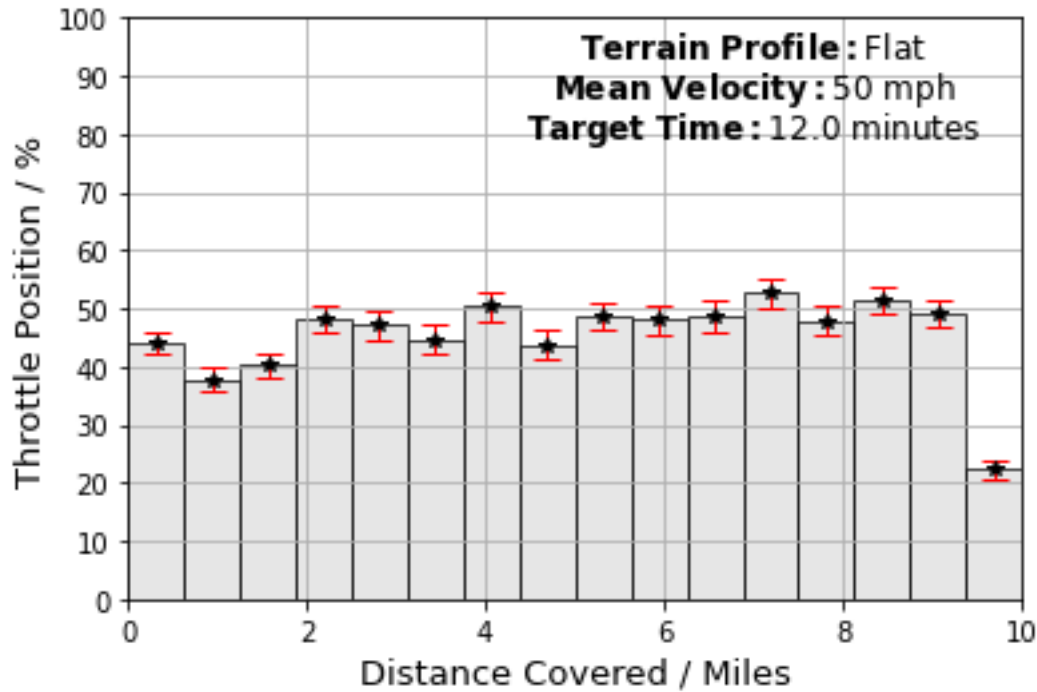


Figure 14: Plot of optimal throttle variation for a car travelling along the 'Flat', 10 mile road, split into 16 segments, at an average speed of 50 mph.

Figures 12-14 show the number of segments chosen were 4, 8 and 16. The error on the solution appears to increase linearly with the number of segments, however further analysis would have to be carried out to determine how the error relates to the number of segments. Given the time constraints of this project, the number of segments have been kept to 8 for the remainder of the collected results in the following sections. This decision was made as this number provided a good enough 'definition' in order to distinguish future trends in driving strategies, but did not compromise the precision of the solutions with larger errors.

4.3 Benchmark Test: Average Speeds

This section aims to showcase the optimal driving strategies for the car travelling along the benchmark profile at varying average speeds and highlight any trends that are found.

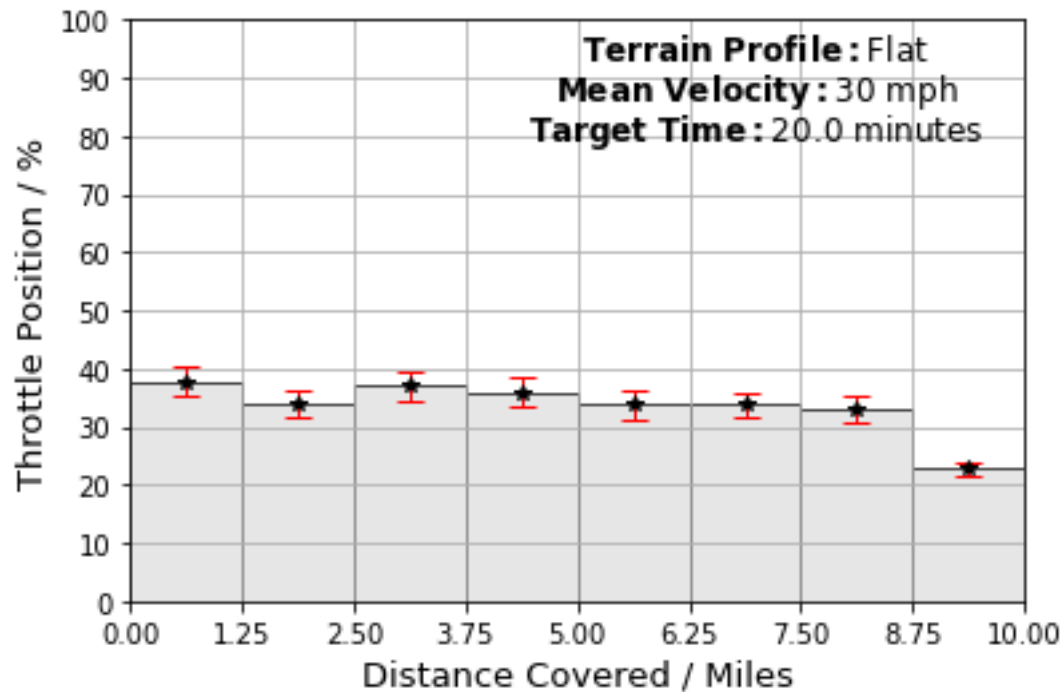


Figure 15: Plot of optimal throttle variation for a car travelling along the 'Flat', 10 mile road, split into 8 segments, at an average speed of 30 mph.

For a car travelling at an average speed of 30 mph along the benchmark 'Flat', 10 mile road (now split into 8 throttle segments), the optimal throttle variation shown in Figure 15 indicates that the most energy efficient method is to keep a constant throttle position until the last segment, dropping the amount of throttle slightly.

The optimal throttle variation shown already in Figure 13 for the same car travelling along the same road profile but now at an average of 50 mph, indicates that it is more energy efficient to steadily increase the throttle very slightly, and then towards the last section or segment of the road, come off the throttle a little bit.

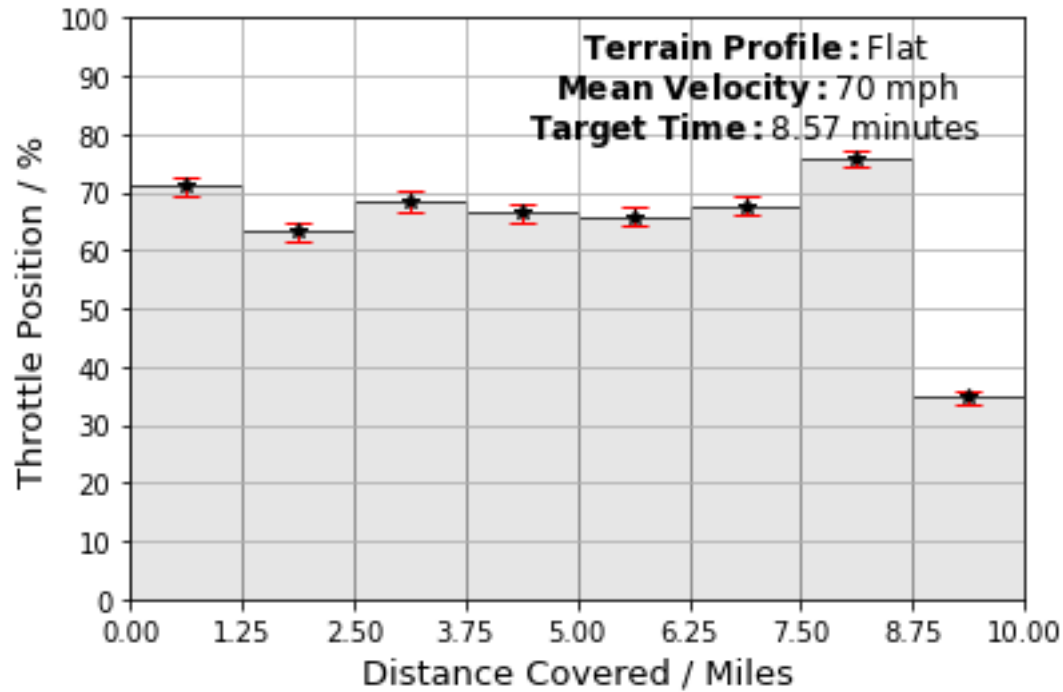


Figure 16: Plot of optimal throttle variation for a car travelling along the 'Flat', 10 mile road, split into 8 segments, at an average speed of 70 mph.

Now for the 70 mph case, results shown in Figure 16 are similar to the 30 mph case in Figure 15, however with an almost double amount of steady throttle, but now reducing the throttle to around half the average previous throttle amount in the final segment of the journey.

These results support the general notion that in order to increase speed, the driver must increase the load on the throttle by a similar proportion. The main trend to take away here is that of the reducing throttle load towards the end of the journey by an observable amount. This hints at a main idea that can be described as a hypermiling technique and that is that driver should make use of the car's inertial motion to save on energy and therefore fuel when the end of journey is know or can be 'observed'.

4.4 Road Length & Terrain Profiles

This section introduces changes in the road profile, from a decrease in length to simulate short journeys, to a variation in the defined terrain profiles described in Section 2.3, and how they affect the optimal driving strategy. The average speed of journeys have been kept to 50 mph for simplicity's sake and to isolate the effect of varying only the road profiles.

4.4.1 Flat

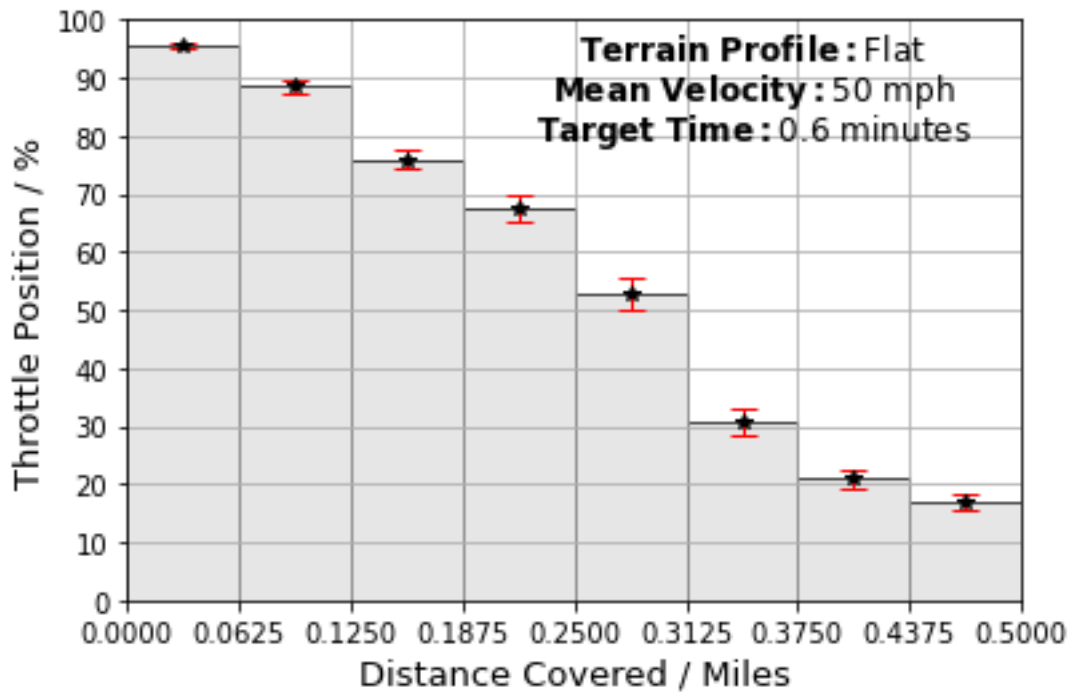


Figure 17: Plot of optimal throttle variation for a car travelling along the 'Flat', half mile road, split into 8 segments, at an average speed of 50 mph.

Comparing Figure 17 with Figure 13, a big difference can be observed to the overall shape of the throttle variation. Figure 17 suggests that when journeys are of shorter distance (half a mile in this case), the most energy efficient way to drive a car is to almost reach full throttle at the beginning of the journey and somewhat linearly reduce the throttle towards the end, however never letting the throttle reach zero.

This could potentially be observed as a magnified version of the last couple of sections in Figure 13, however, a higher definition is required to confirm this and hence a larger number of segments coupled with a larger number of journey simulations (larger than 300,000 to reduce errors and solution instability).

4.4.2 Incline

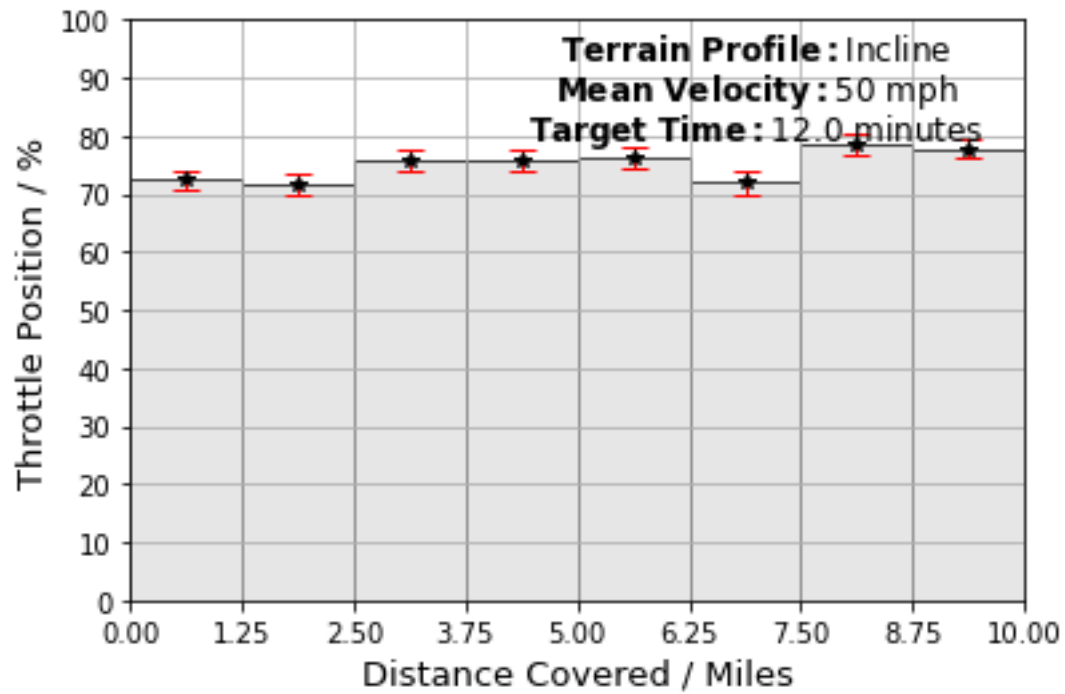


Figure 18: Plot of optimal throttle variation for a car travelling along the 'Incline', 10 mile road, split into 8 segments, at an average speed of 50 mph.

Observing Figure 18 for the long 10 mile journey on the 'Incline' road, the optimal driving strategy to travel at an average 50 mph seems to be to keep throttle to a high average of 75% for the entire journey, a rather simple case.

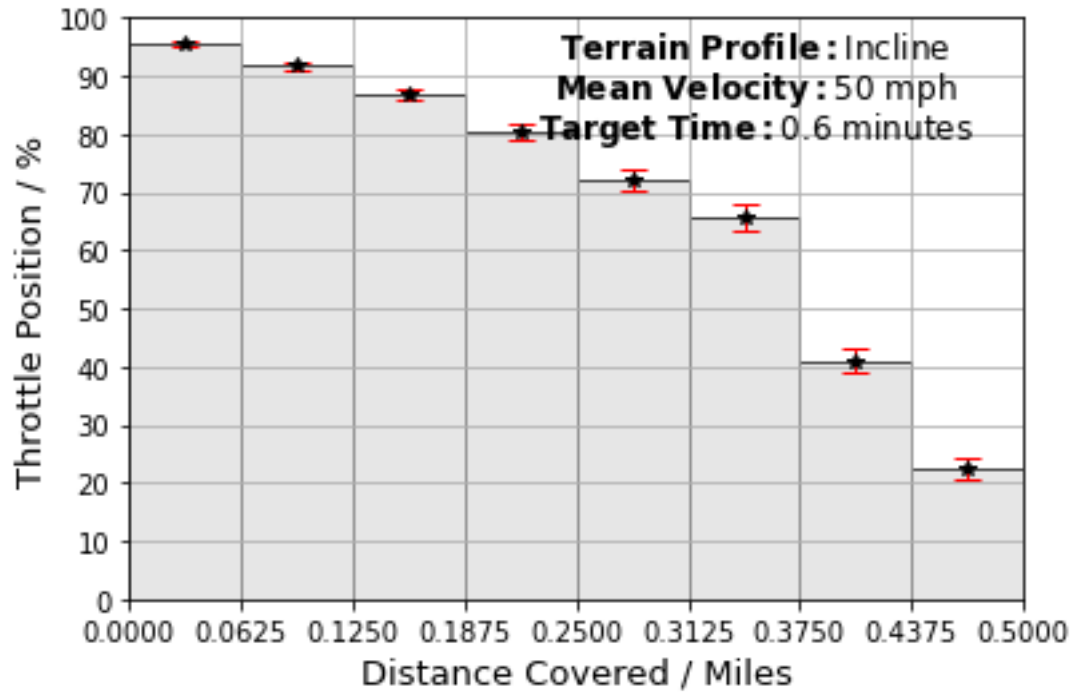


Figure 19: Plot of optimal throttle variation for a car travelling along the 'Incline', half mile road, split into 8 segments, at an average speed of 50 mph.

For the short, half mile journey however, the more energy efficient method is similar to that of the short 'Flat' journey, to start of at near full throttle and ease off the throttle as the journey progresses, however to an overall lesser degree.

4.4.3 Decline

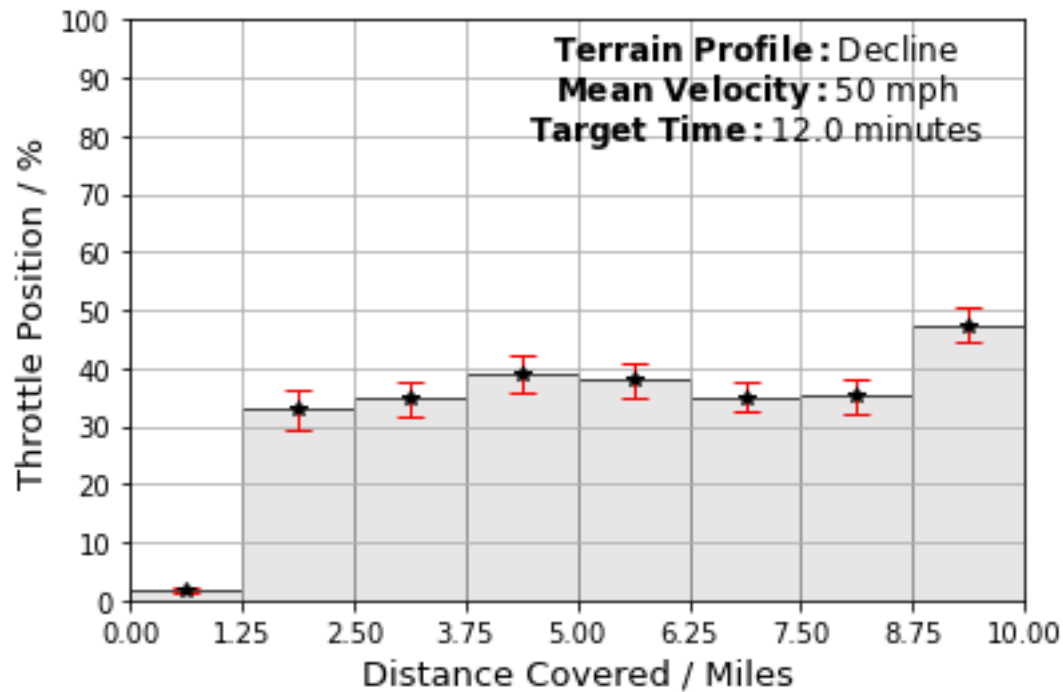


Figure 20: Plot of optimal throttle variation for a car travelling along the 'Decline', 10 mile road, split into 8 segments, at an average speed of 50 mph.

Now analysing the results for the 10 mile, 'Decline' terrain profile, it can be seen that the more energy efficient way to drive the car is to barely apply any throttle to the engine for the first segment, most likely indicating that energy would be saved when relying on the weight of the car acting downwards along the slope to propel it in the forward direction. This is a very intuitive result which is expected, as this way the driver is making use of external forces such as gravity to get the car moving forward. For the rest of the journey the optimal strategy seems to be keep an overall constant throttle on the engine.

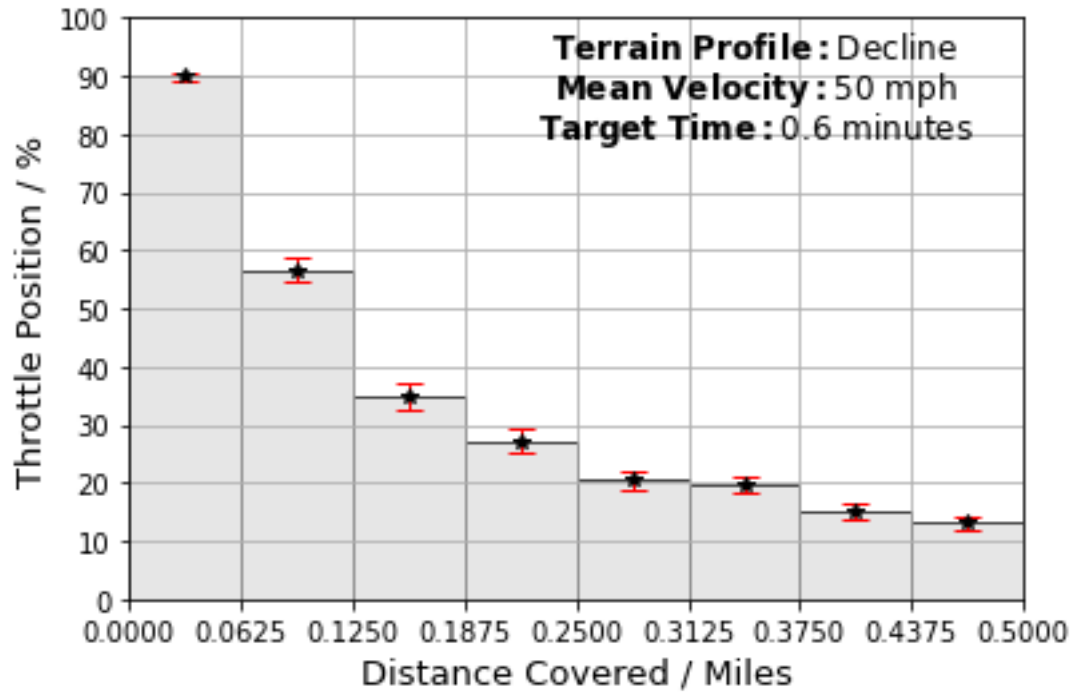


Figure 21: Plot of optimal throttle variation for a car travelling along the 'Decline', half mile road, split into 8 segments, at an average speed of 50 mph.

For a short journey however, the result appears to oppose that for a long journey along the same 'Decline' road as for this case the results suggest to accelerate quickly with around 90% throttle and decrease the throttle as the car progresses along the road, but never reducing to zero, staying above about 10% always.

This is likely to mean that in order to achieve an average speed of 50 mph in a shorter distance, harder acceleration is needed compared to a long journey, as in the case for the 10 mile journey, the car has a longer amount of time and journey length to achieve the same average speed, and so makes use of the gravitational force to get the car into motion when resistive forces are large enough already.

4.4.4 Flat Top

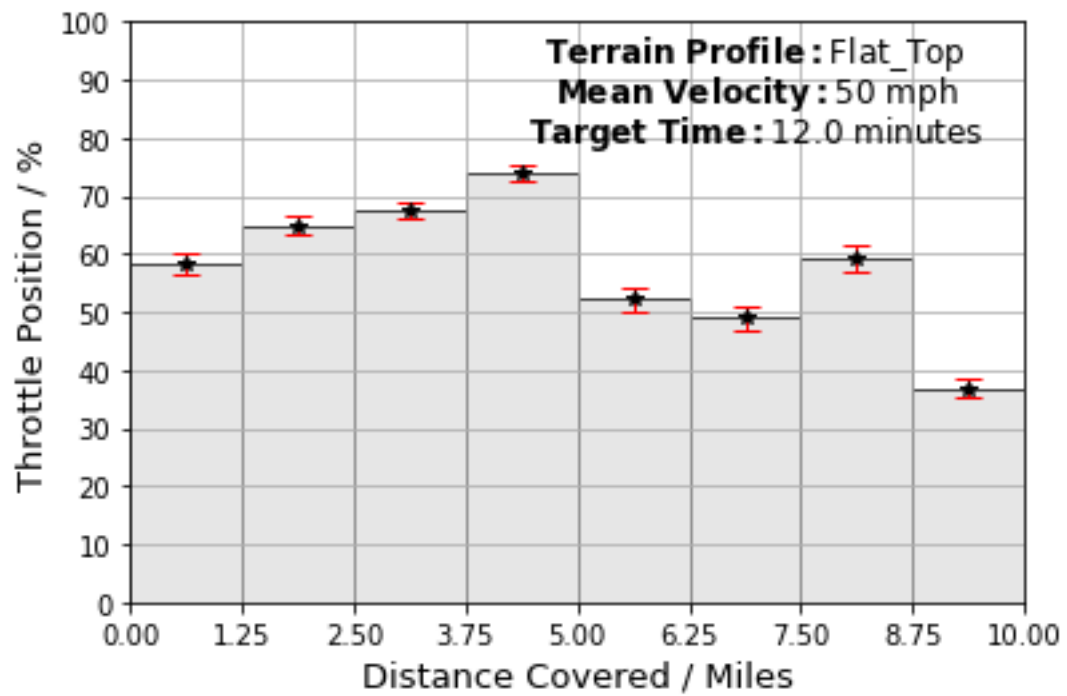


Figure 22: Plot of optimal throttle variation for a car travelling along the 'Flat Top', 10 mile road, split into 8 segments, at an average speed of 50 mph.

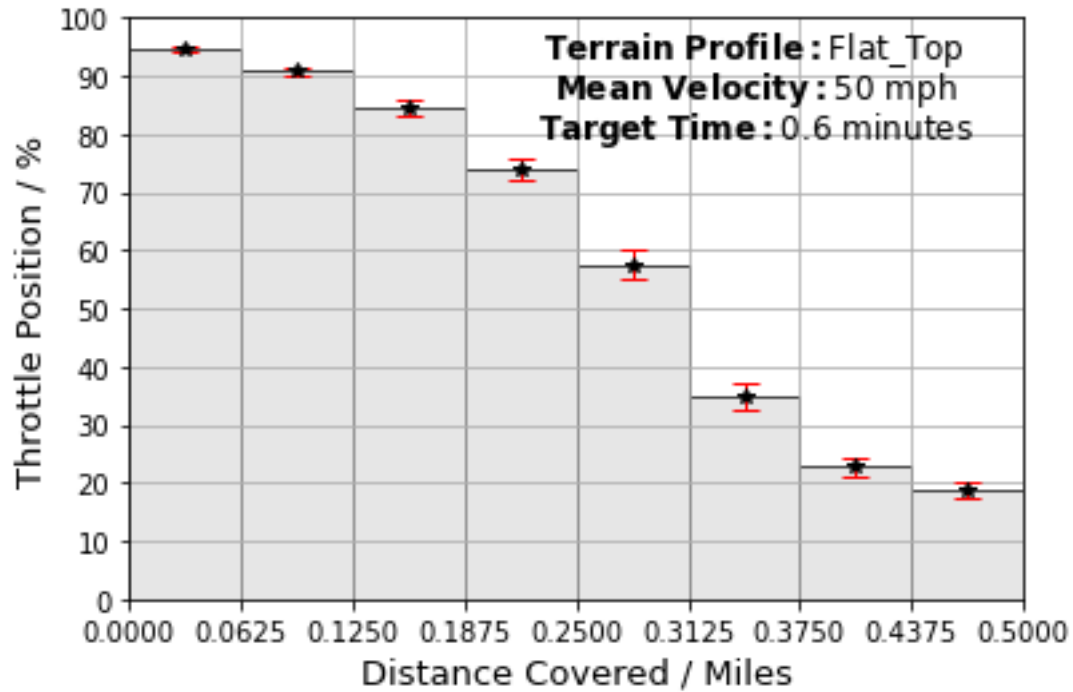


Figure 23: Plot of optimal throttle variation for a car travelling along the 'Flat Top', half mile road, split into 8 segments, at an average speed of 50 mph.

The results for the 'Flat Top' profile may require further analysis with higher resolution (more segments and more simulations) particularly for Figure 22, as it is expected that this journey, being comprised of a 'Flat' section for 5 miles and then 'Incline' for another 5 miles, would be equivalent to 'Flat' and 'Incline' throttle optimal variations from Figures 13 and 18. However, although the first half of Figure 22 does follow this expectation, the second half may need to be examined in higher resolution to see how this optimal strategy varies from a basic composite of the two profiles.

The half mile variation in Figure 23, seems to agree with the composite of Figures 17 and 19 though, seeing as they both follow a similar pattern on their own.

4.4.5 Flat Bottom

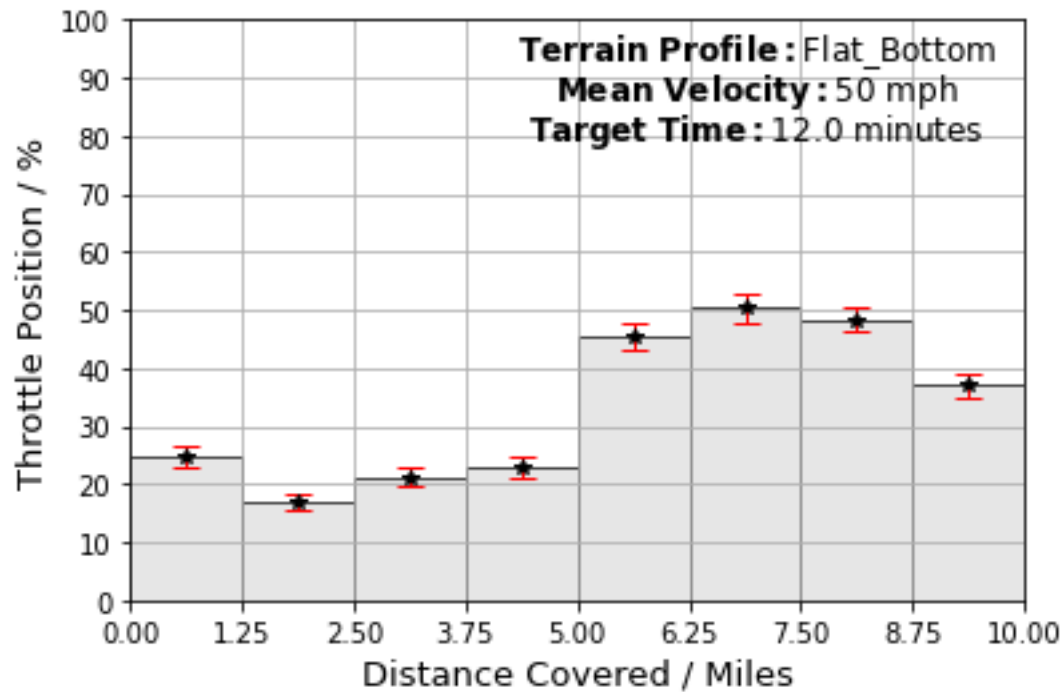


Figure 24: Plot of optimal throttle variation for a car travelling along the 'Flat Bottom', 10 mile road, split into 8 segments, at an average speed of 50 mph.

The resulting throttle plot for the 'Flat Bottom' long journey is curious, as it doesn't seem to follow the 'Flat' variation for the first half of the journey, and instead appears to start off around 25% and reduce the throttle by an observable amount, before increasing to around double when reaching the 'Decline' section. This diversion from the simple composite of the two individual throttle optimal solution may be a result of the simple fact that if the system or driver 'knows' or can 'see' the 'Decline' section coming up ahead, the most energy efficient strategy is to keep acceleration to a minimum and simply go faster on the downhill 'Decline' section where 'free' kinetic energy from gravity can be made use of instead.

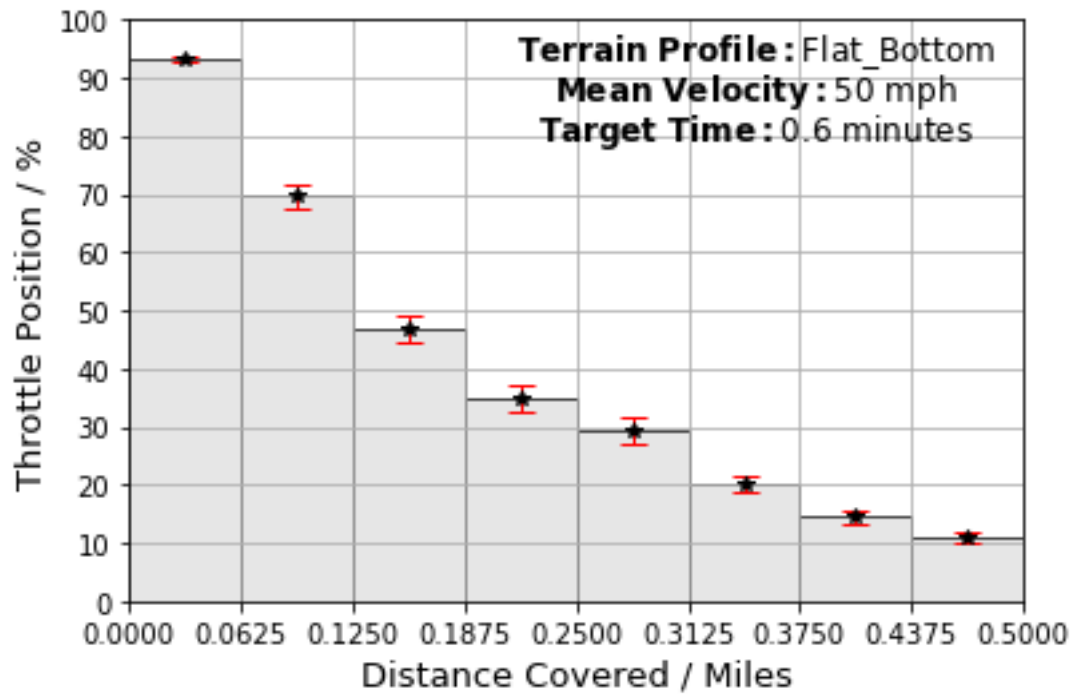


Figure 25: Plot of optimal throttle variation for a car travelling along the 'Flat Bottom', half mile road, split into 8 segments, at an average speed of 50 mph.

As expected with the 'Flat Top' case, the short half mile journey along the 'Flat Bottom' profile can be observed to have an optimal driving strategy to accelerate hard in the beginning and reduce load on throttle roughly linearly with distance. This makes sense seeing as the optimal variations for short 'Flat' and 'Decline' profiles are similar in shape.

4.4.6 Bumpy

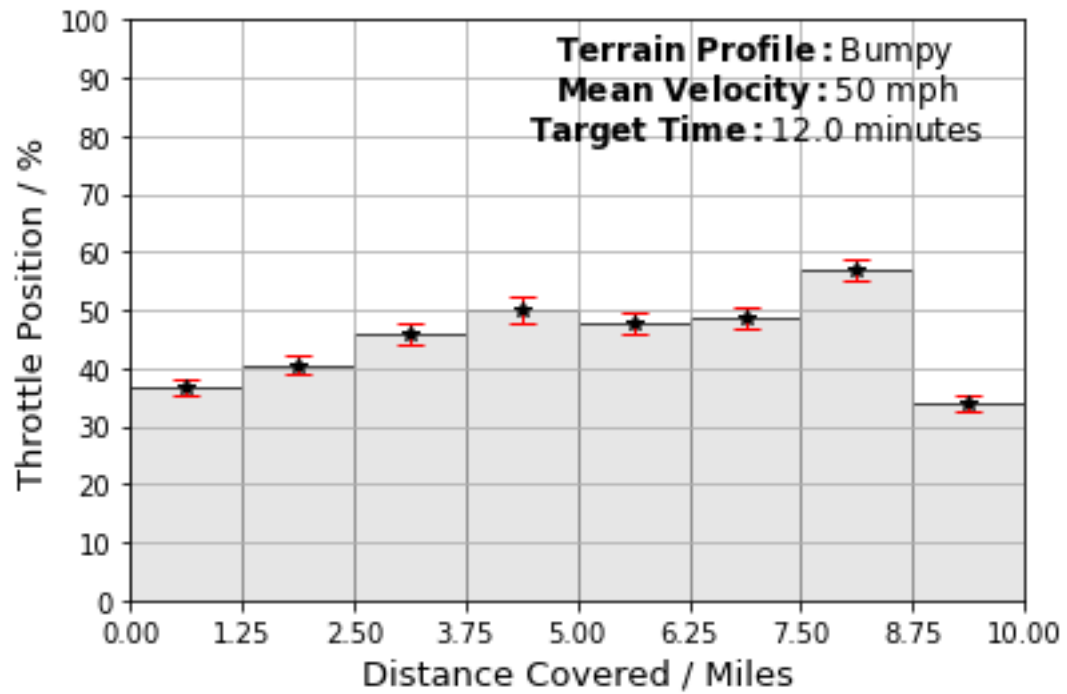


Figure 26: Plot of optimal throttle variation for a car travelling along the 'Bumpy', 10 mile road, split into 8 segments, at an average speed of 50 mph.

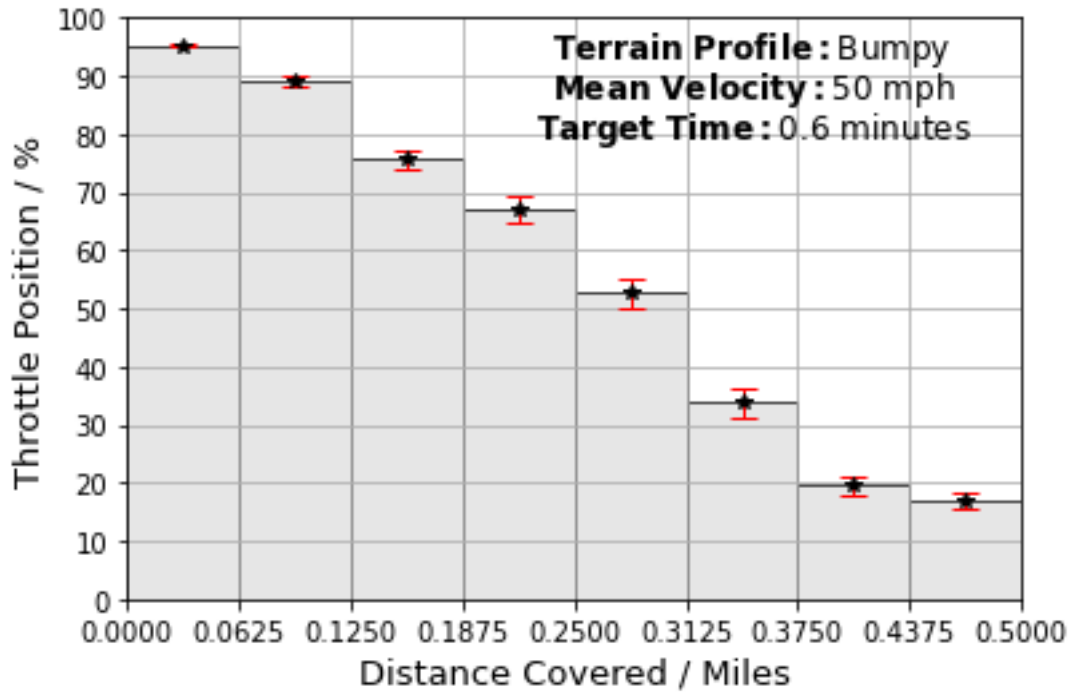


Figure 27: Plot of optimal throttle variation for a car travelling along the 'Bumpy', half mile road, split into 8 segments, at an average speed of 50 mph.

Finally, the 'Bumpy' (or realistic road with irregularities along the road of randomised slopes between -3° and 3°) profile produces very similar results to that for the 'Flat' profile, with very little variation. This is likely due to the fact that the randomised slopes follow a random sampling distribution centred around 0, however this means that slight variations in angles along the road don't play a huge factor on how best to drive to save energy and fuel. Note that the results still indicate to reduce throttle substantially in the final segment for the long journey given in Figure 28, making use of built up momentum to drive the car forward for the last couple of miles of the journey.

5 Conclusion

So, what is the most efficient way to drive a car? Well, if time is of no concern to the driver, then the best way to save on fuel is to drive very slowly and with just enough throttle to overcome resistive forces. However, practically the answer depends on how the driver wishes to balance their energy used by the engine for the journey and how long they expect the journey should take. This project was carried out in order to settle this debate on which ways are the most efficient, by making use of an analytical model coded into Python from scratch to meet best meet the requirements of the project.

It seems as though the most efficient way to drive a car is to use the vehicle's own inertia to carry the car forward either when enough momentum has been built up and a certain distance of the journey is left, or when the terrain works in the car's favour, as with any downhill journeys. An important thing to note as well is that for composite journeys, where both flat sections and uphill sections exist, the energy efficient approach is not always the composite of the two individual strategies back to back. Instead, it is more efficient to know what is coming up ahead and adjust the approach accordingly, always trying to make use of the vehicle's momentum where possible.

Note that the driving strategies determined in this project should not serve as the only guide to those

that want to apply them in their every day driving habits, as they do not take traffic or other hazardous scenarios into account, and so these may not be the safest approach in real driving situations.

This project targets a very layered question, and although a significant part of it has been answered, it requires further investigation in order to determine every possible aspect of the most efficient strategies to encompass all driving scenarios.

References

- [1] Buchanan, R.A. (1998). *"History of Technology: Internal Combustion engines"*. Encyclopedia Britannica.
<https://doi.org/10.1098/rspa.1931.0162>
- [2] Zal, P. (2020). *"2020 Ford Fiesta 1.0 EcoBoost (125) (man. 6) (Titanium) (model up to June 2020 for Europe) car specifications & performance data review"*.
- [3] The Engineering Toolbox. *"Rolling Resistance"*.
<https://doi.org/10.1098/rspa.1931.0162>
- [4] The Engineering Toolbox. *"Air - Density, Specific Weight and Thermal Expansion Coefficient vs. Temperature and Pressure"*.
<https://doi.org/10.1098/rspa.1931.0162>
- [5] Ni, D. (2016). *"Traffic Flow Theory, Chapter 19: Engine Modeling"*. Butterworth-Heinemann. 9780128041475.
<https://books.google.co.uk/books?id=LVMrCQAAQBAJ>
- [6] Motortrend. *"Where'd My Horsepower Go? Drivetrain Power Loss & The 15% "Rule"'*.
<https://www.motortrend.com/how-to/modp-1005-drivetrain-power-loss/>

Appendix

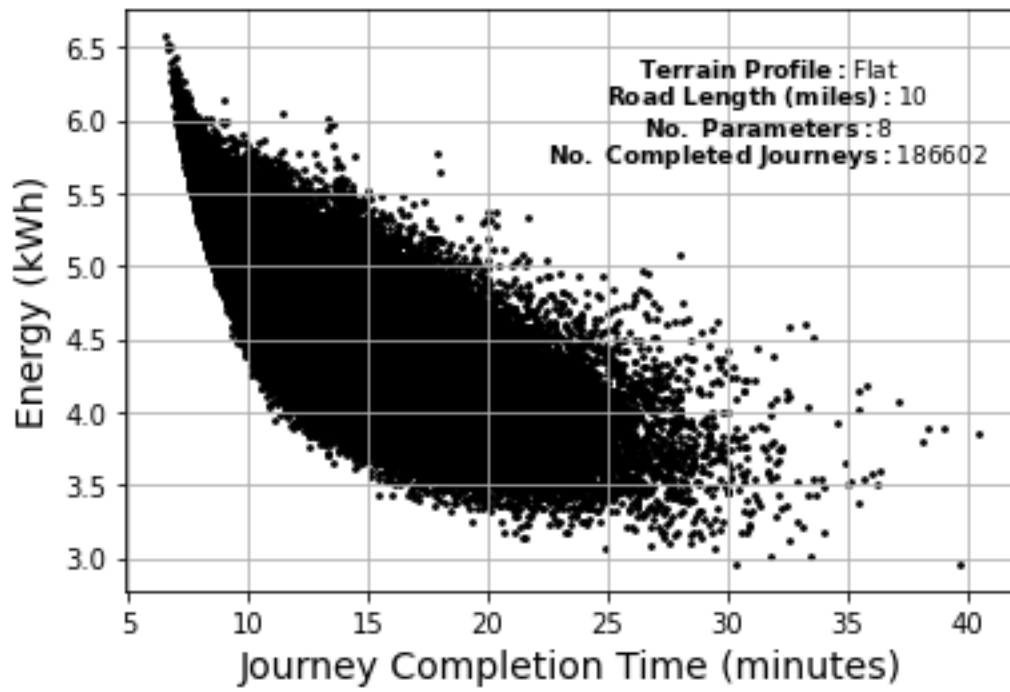


Figure 28: Scatter plot of energy (in kWh) against completion time for benchmark road profile ('Flat', 10 mile road split into 8 segments).

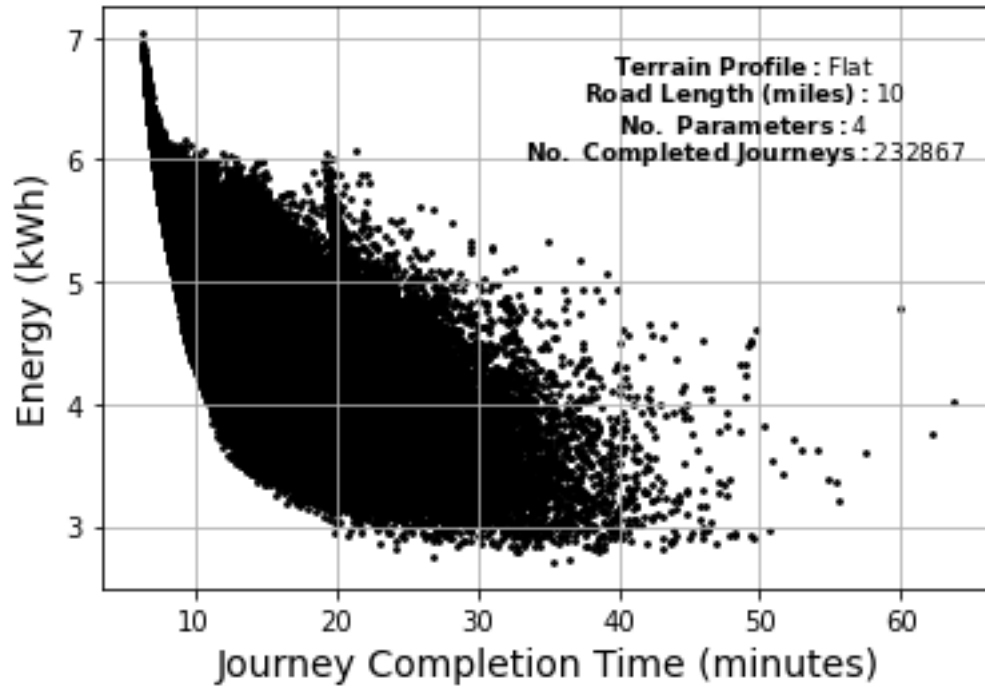


Figure 29: Scatter plot of energy (in kWh) against completion time for 'Flat', 10 mile road split into 4 segments.

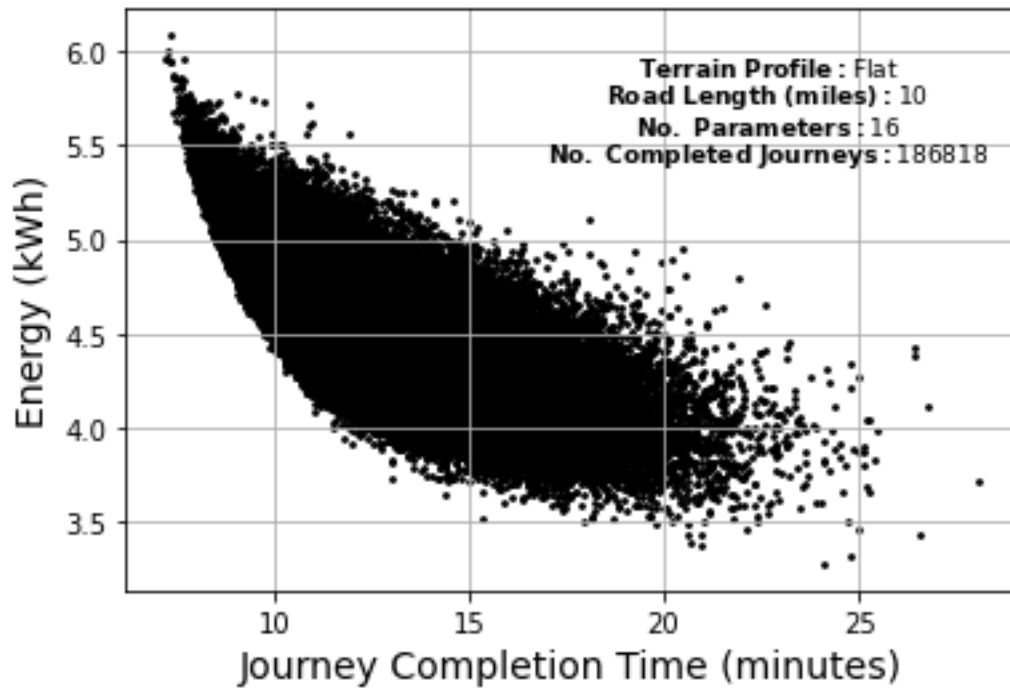


Figure 30: Scatter plot of energy (in kWh) against completion time for 'Flat', 10 mile road split into 16 segments.

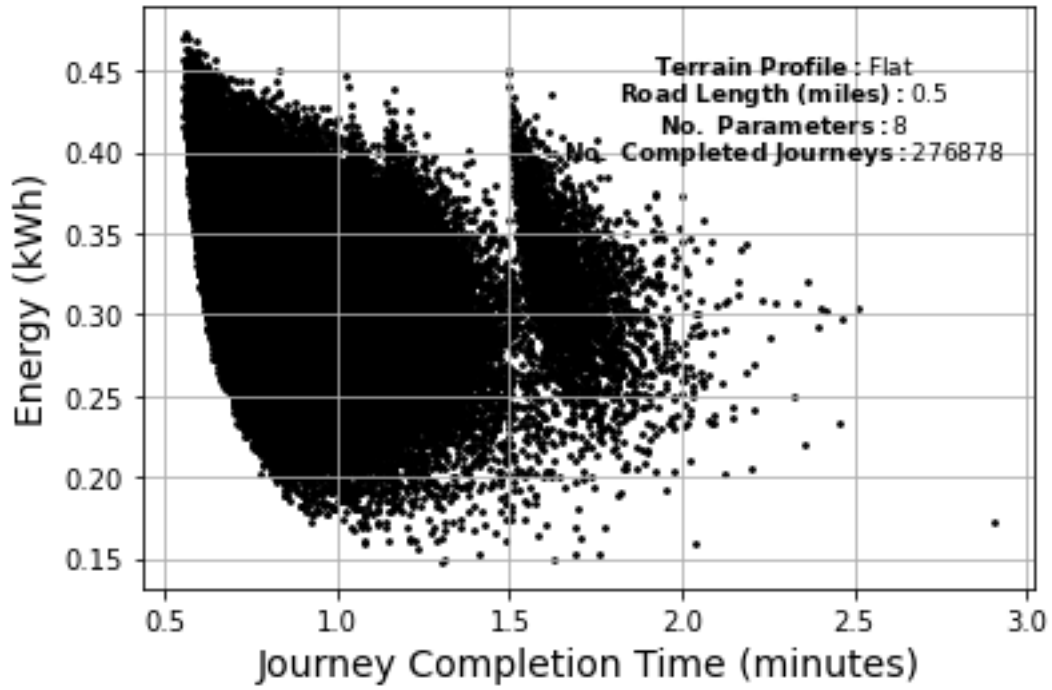


Figure 31: Scatter plot of energy (in kWh) against completion time for 'Flat', half mile road split into 8 segments.

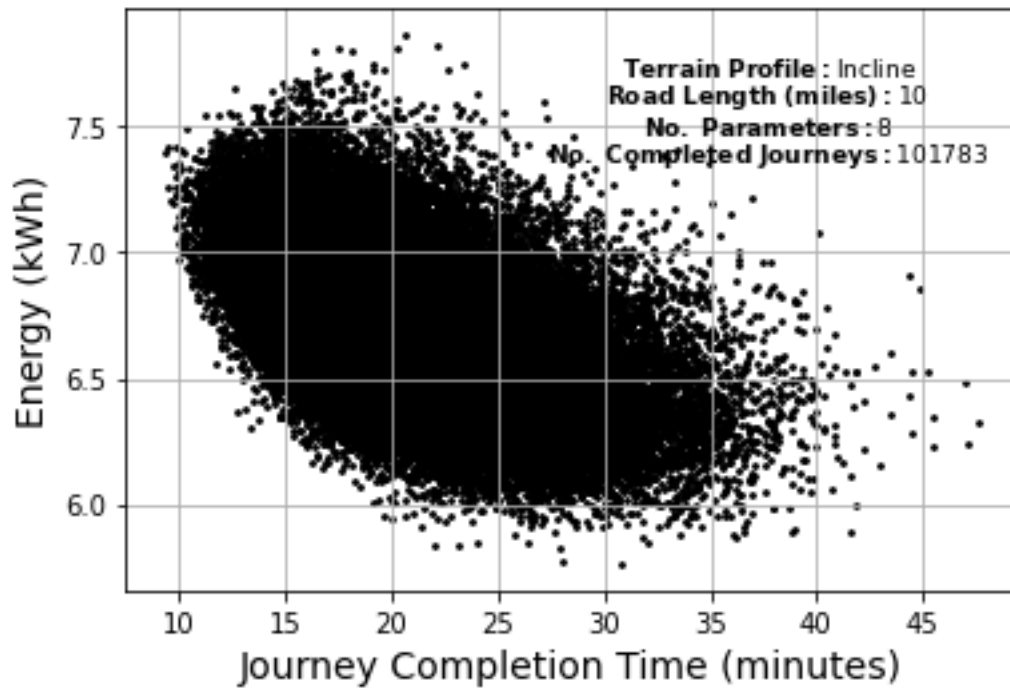


Figure 32: Scatter plot of energy (in kWh) against completion time for 'Incline', 10 mile road split into 8 segments.

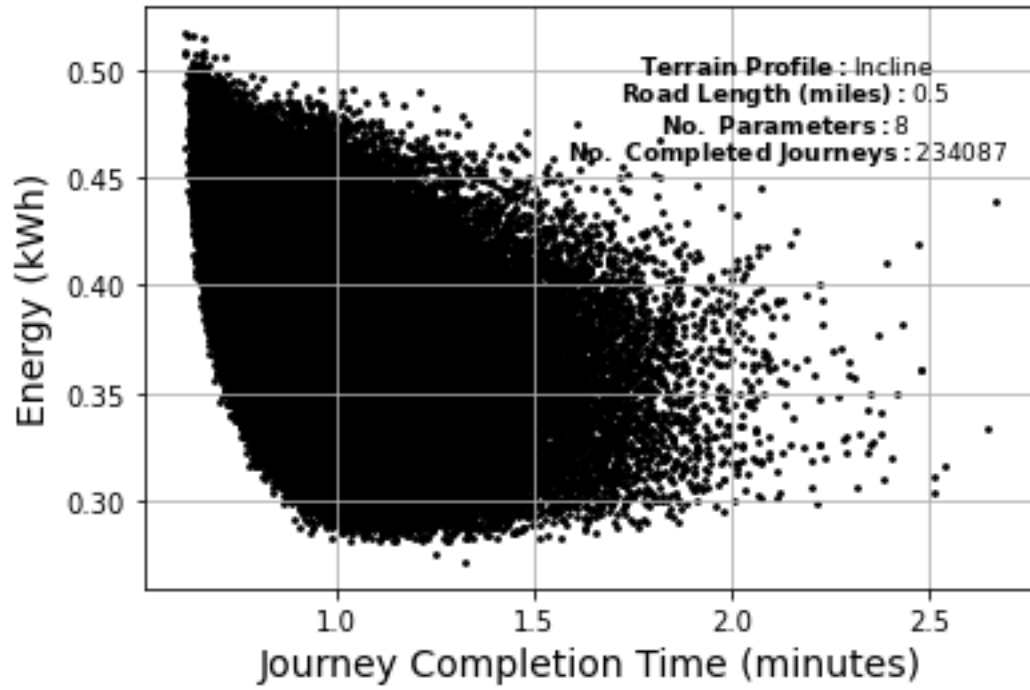


Figure 33: Scatter plot of energy (in kWh) against completion time for 'Incline', half mile road split into 8 segments.

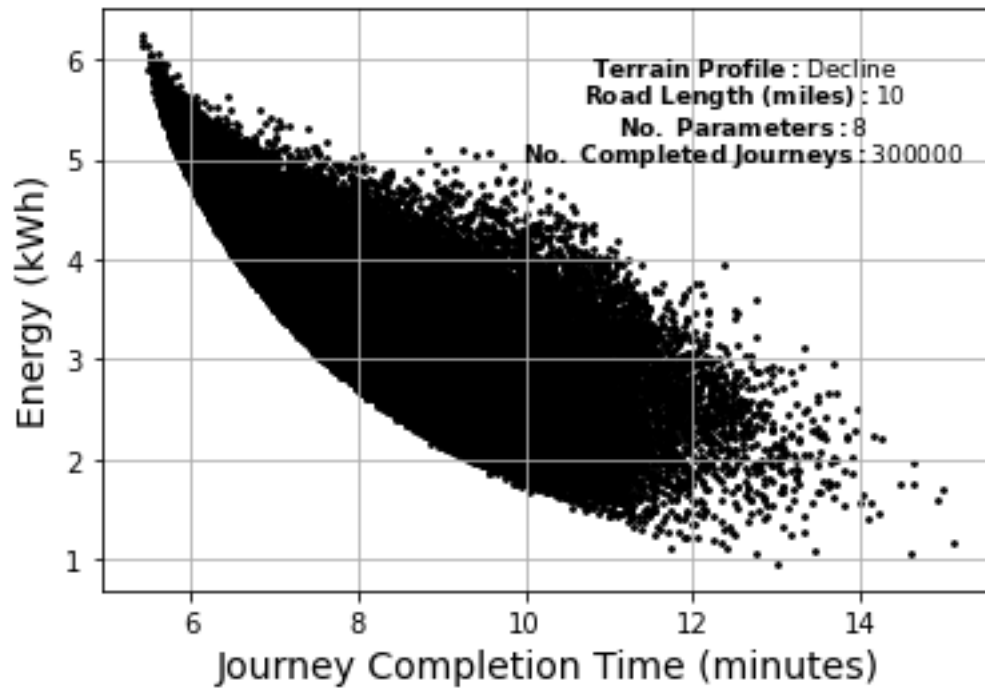


Figure 34: Scatter plot of energy (in kWh) against completion time for 'Decline', 10 mile road split into 8 segments.

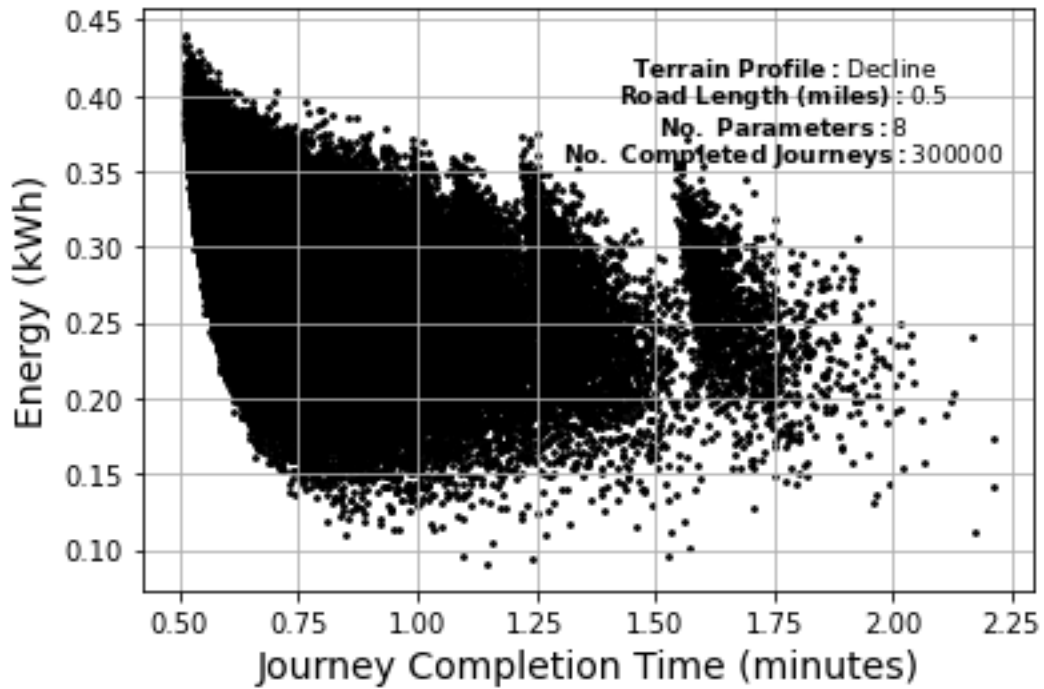


Figure 35: Scatter plot of energy (in kWh) against completion time for 'Decline', half mile road split into 8 segments.

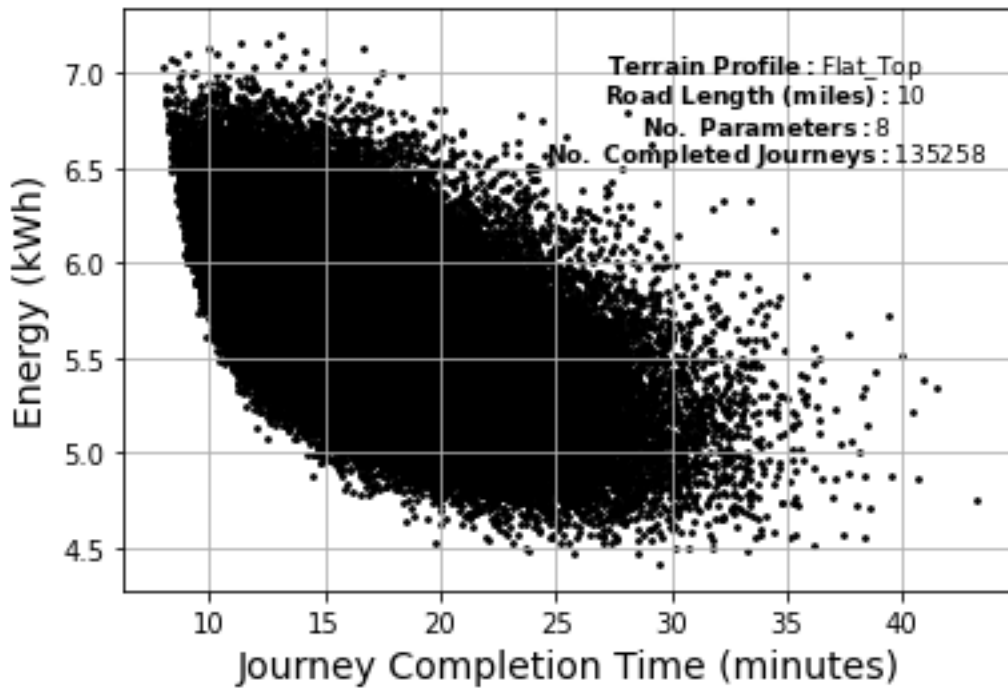


Figure 36: Scatter plot of energy (in kWh) against completion time for 'Flat Top', 10 mile road split into 8 segments.

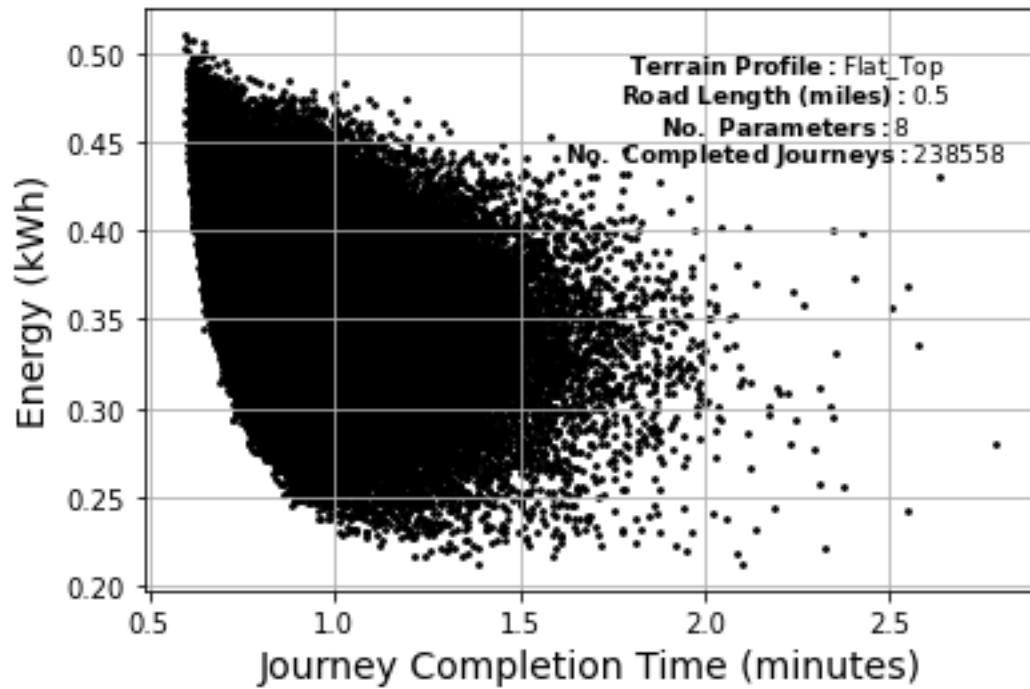


Figure 37: Scatter plot of energy (in kWh) against completion time for 'Flat Top', half mile road split into 8 segments.

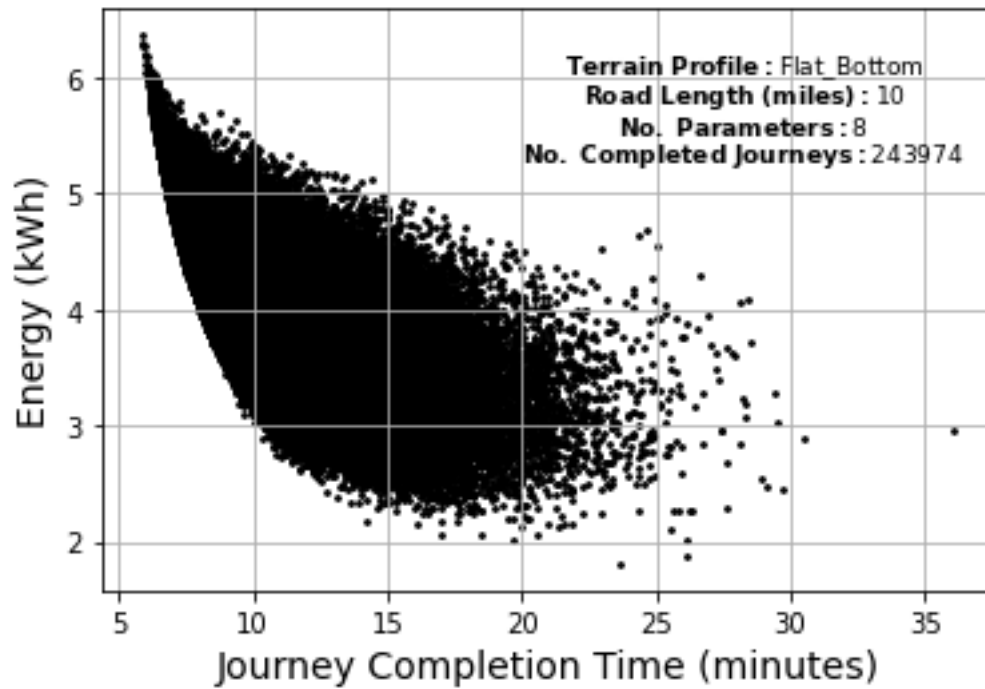


Figure 38: Scatter plot of energy (in kWh) against completion time for 'Flat Bottom', 10 mile road split into 8 segments.

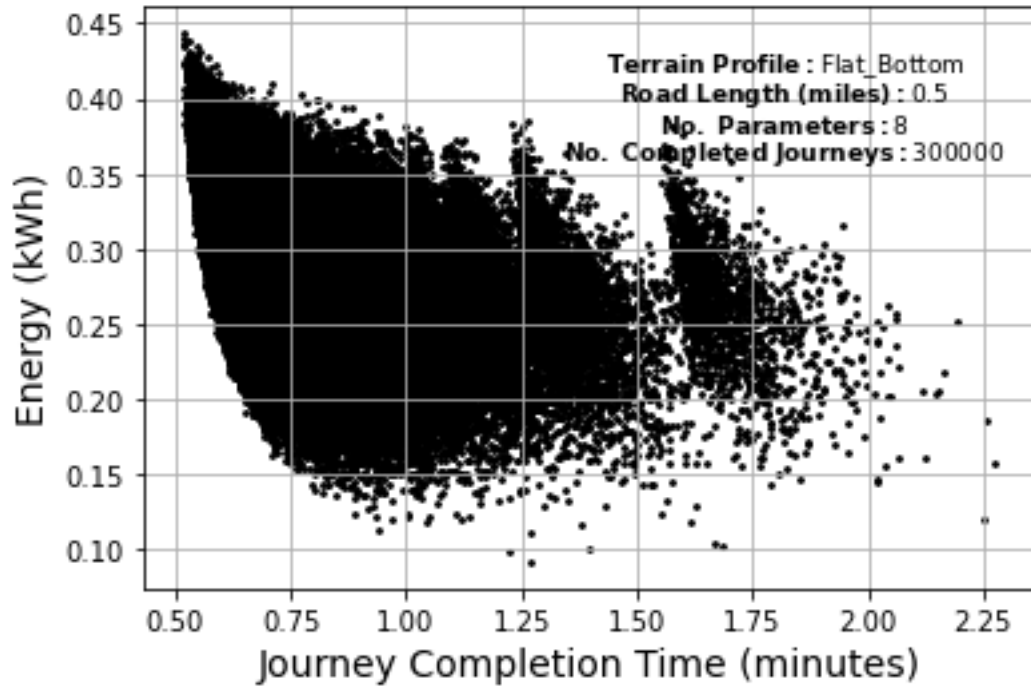


Figure 39: Scatter plot of energy (in kWh) against completion time for 'Flat Bottom', half mile road split into 8 segments.

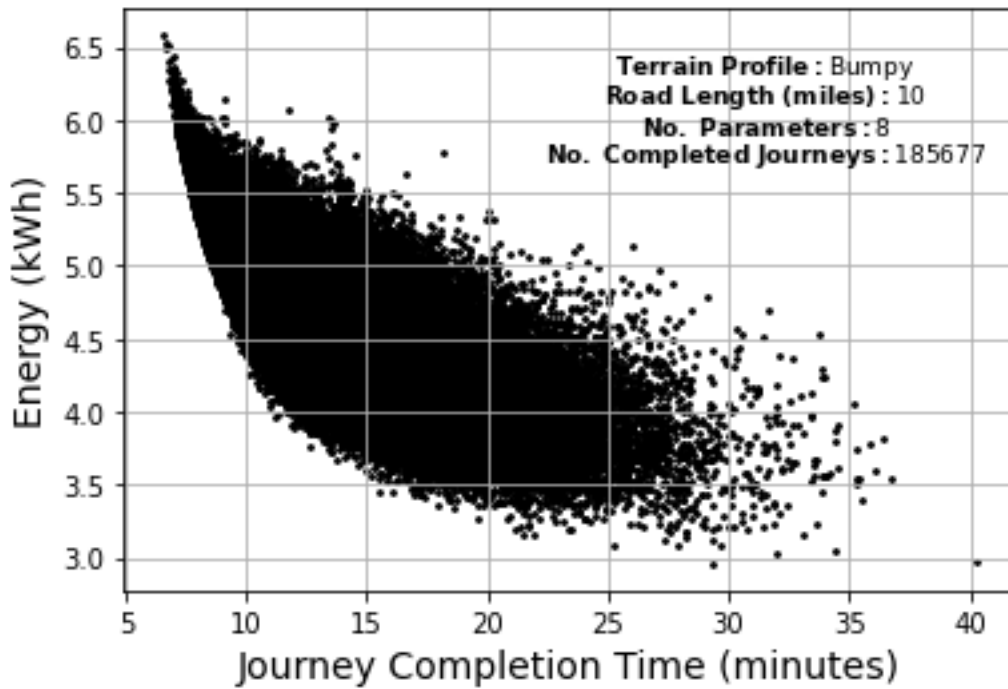


Figure 40: Scatter plot of energy (in kWh) against completion time for 'Bumpy', 10 mile road split into 8 segments.

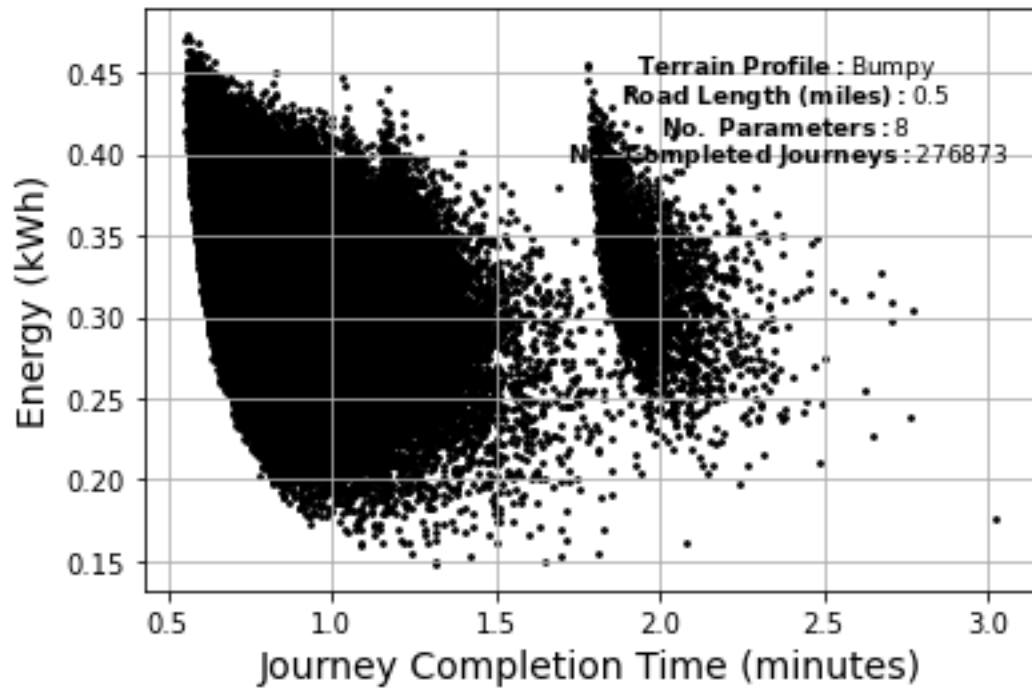


Figure 41: Scatter plot of energy (in kWh) against completion time for 'Bumpy', half mile road split into 8 segments.